

ESAT PRACTICE TEST SERIES

MATHS 2 + BIOLOGY

LEVEL 2 • HARDER THAN ESAT

Maths 2 + Biology

Mathematics 1 + Mathematics 2 + Biology

Challenge material with denser interpretation, quantitative reasoning and longer solution chains.

MODULE

Mathematics 1

MODULE

Mathematics 2

MODULE

Biology

FULL SOLUTION BOOK

ThrivingScholars

81 preserved questions • three verified module keys

Maths 2 + Biology · Level 2 Harder than ESAT

This book compiles three existing 27-question modules into one 81-question pathway. No new questions have been generated and no source solution has been rewritten.

Composition: Mathematics 1 + Mathematics 2 + Biology. Each module retains its original question order, answer and worked solution.

Mathematics 1

Engineering Mock Test 5 · preserved Mathematics 1 module

Q1 C	Q2 B	Q3 D	Q4 A	Q5 E
Q6 C	Q7 B	Q8 D	Q9 A	Q10 E
Q11 C	Q12 B	Q13 D	Q14 A	Q15 E
Q16 B	Q17 C	Q18 D	Q19 A	Q20 E
Q21 C	Q22 B	Q23 D	Q24 A	Q25 E
Q26 C	Q27 B			

Mathematics 2

Engineering Mock Test 5 · preserved Mathematics 2 module

Q1 D	Q2 B	Q3 E	Q4 C	Q5 B
Q6 C	Q7 B	Q8 B	Q9 C	Q10 B
Q11 B	Q12 B	Q13 C	Q14 B	Q15 B
Q16 C	Q17 C	Q18 D	Q19 D	Q20 C
Q21 C	Q22 B	Q23 C	Q24 C	Q25 B
Q26 E	Q27 D			

Biology

Preserved Biology Harder than ESAT source set

Q1 C	Q2 F	Q3 B	Q4 C	Q5 C
Q6 D	Q7 C	Q8 D	Q9 C	Q10 B
Q11 B	Q12 E	Q13 F	Q14 E	Q15 C
Q16 D	Q17 B	Q18 B	Q19 E	Q20 E
Q21 A	Q22 E	Q23 D	Q24 F	Q25 E
Q26 C	Q27 H			

Mathematics 1

27 QUESTIONS • COMPLETE SOLUTIONS

Engineering Mock Test 5 · preserved Mathematics 1 module. Question wording, option order, answer and solution method are unchanged.

M1-01

Laws of indices and exact simplification

Answer C**QUESTION 1**

Simplify fully

$$\frac{(50x^{3/2}y^{-1})^2(2x^{-1}y^{5/2})^3}{(20x^{1/2}y)^2},$$

where x and y are positive.

A $\frac{10y^3\sqrt{y}}{x}$

B $\frac{50y^3}{x}$

C $\frac{50y^3\sqrt{y}}{x}$

D $\frac{50\sqrt{y}}{x}$

E $\frac{250y^3\sqrt{y}}{x}$

WORKED SOLUTION

$$(50x^{3/2}y^{-1})^2 = 2500x^3y^{-2} \text{ and } (2x^{-1}y^{5/2})^3 = 8x^{-3}y^{15/2}.$$

So the numerator is $2500x^3y^{-2} \cdot 8x^{-3}y^{15/2} = 20000y^{11/2}$.

The denominator is $(20x^{1/2}y)^2 = 400xy^2$.

Therefore

$$\frac{20000y^{11/2}}{400xy^2} = \frac{50y^{7/2}}{x} = \boxed{\frac{50y^3\sqrt{y}}{x}}.$$

M1-02

Successive percentage change and comparison

Answer B

QUESTION 2

The price of item P is reduced by 20%, and then a sales tax of 5% is applied to the reduced price.

The price of item Q is increased by 10%, and then reduced by 10%.

After these two-step changes, the final prices of P and Q are equal.

What was the ratio of the original price of P to the original price of Q?

A 28 : 33

B 33 : 28

C 11 : 10

D 6 : 5

E 99 : 80

WORKED SOLUTION

Final price of P is $0.8 \times 1.05P = 0.84P$.

Final price of Q is $1.1 \times 0.9Q = 0.99Q$.

So $0.84P = 0.99Q$, giving

$$\frac{P}{Q} = \frac{0.99}{0.84} = \frac{99}{84} = \frac{33}{28}.$$

M1-03

Algebraic fractions and hidden substitution

Answer D

QUESTION 3

For real x , define

$$E = \frac{\frac{1}{x-3} - \frac{1}{x+2}}{\frac{1}{x-3} + \frac{1}{x+2}}.$$

Given that $x > 3$ and

$$E + \frac{1}{E} = \frac{13}{2},$$

what is the value of x ?

A $\frac{11}{2}$

B 8

C 12

D $\frac{31}{2}$

E 18

WORKED SOLUTION

Simplify first:

$$E = \frac{\frac{(x+2)-(x-3)}{(x-3)(x+2)}}{\frac{(x+2)+(x-3)}{(x-3)(x+2)}} = \frac{5}{2x-1}.$$

Let $t = E$. Then $t + \frac{1}{t} = \frac{13}{2}$, so

$$2t^2 - 13t + 2 = 0 = (2t - 1)(t - 6).$$

Hence $t = 6$ or $t = \frac{1}{6}$. Since $x > 3$, we have $0 < E < 1$, so $E = \frac{1}{6}$.

Therefore

$$\frac{5}{2x-1} = \frac{1}{6} \Rightarrow 2x - 1 = 30 \Rightarrow \boxed{x = \frac{31}{2}}.$$

M1-04

Inequalities with domain restrictions

Answer A

QUESTION 4

How many integer values of x satisfy all three conditions?

- $-5 \leq x \leq 5$
- $\frac{x^2 + 3x - 10}{x^2 - 1}$ is defined
- $\frac{x^2 + 3x - 10}{x^2 - 1} \leq 1$

A 6

B 5

C 4

D 7

WORKED SOLUTION

Bring everything to one side:

$$\frac{x^2 + 3x - 10}{x^2 - 1} - 1 \leq 0 \Rightarrow \frac{3x - 9}{(x - 1)(x + 1)} \leq 0.$$

So we solve

$$\frac{x - 3}{(x - 1)(x + 1)} \leq 0, \quad x \neq \pm 1.$$

A sign chart gives $(-\infty, -1) \cup (1, 3]$.

Intersecting with $-5 \leq x \leq 5$ gives $-5, -4, -3, -2, 2, 3$, so there are 6 values.

M1-05

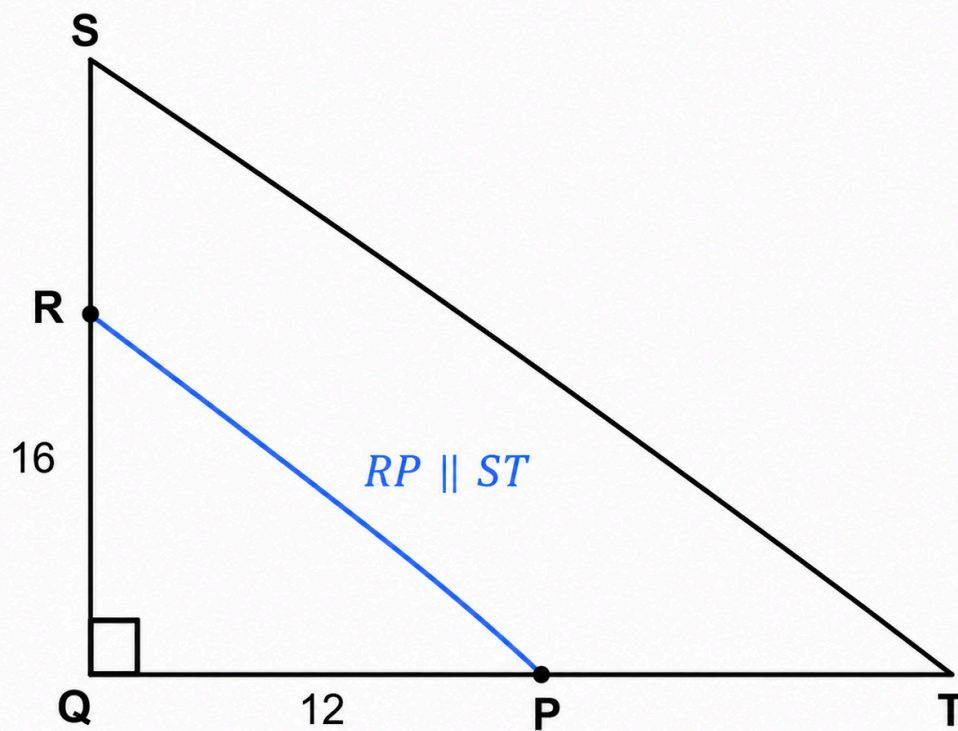
Similarity and area scaling in a right triangle

Answer E

QUESTION 5

Triangle SQT is right-angled at Q, with SQ = 16 cm and QT = 12 cm.

Point R lies on SQ such that SR : RQ = 1 : 3. Through R, a line parallel to ST meets QT at P.



What is the area of quadrilateral SRPT?

A 24 cm^2

B 30 cm^2

C 36 cm^2

D 40 cm^2

E 42 cm^2

WORKED SOLUTION

The area of the whole triangle is $\frac{1}{2} \cdot 16 \cdot 12 = 96$.

Since $SR : RQ = 1 : 3$, we get $RQ = \frac{3}{4} \cdot 16 = 12$.

Because $RP \parallel ST$, triangles QRP and QST are similar with scale factor

$$\frac{QR}{QS} = \frac{12}{16} = \frac{3}{4}.$$

So $QP = \frac{3}{4} \cdot 12 = 9$. Then area of triangle QRP is $\frac{1}{2} \cdot 12 \cdot 9 = 54$.

Hence quadrilateral $SRPT$ has area $96 - 54 = \boxed{42 \text{ cm}^2}$.

M1-06

Arithmetic sequences and common terms

Answer **C**

QUESTION 6

The n th term of sequence T is $4n - 3$.

The n th term of sequence V is $7n + p$, where p is an integer and $0 \leq p < 7$.

The sequences have consecutive common terms 53 and 81.

What is the value of p ?

A 1

B 2

C 4

D 5

E 6

WORKED SOLUTION

Using the common term 53,

$$53 = 7n + p.$$

Since $53 \equiv 4 \pmod{7}$, and $0 \leq p < 7$, we must have $p = 4$.

M1-07

Cylinders with swapped dimensions

Answer **B**

QUESTION 7

Two solid cylinders P and Q are defined using positive numbers x and y , where $x > y$.

- Cylinder P has diameter x and height y .
- Cylinder Q has diameter y and height x .

The total surface area of P is $\frac{8}{5}$ of the total surface area of Q .

The volume of P exceeds the volume of Q by $32\pi \text{ cm}^3$.

What is the value of $x + y$?

A 10

B 12

C 14

D 16

E 18

WORKED SOLUTION

For a cylinder of diameter d and height h , total surface area is

$$2\pi\left(\frac{d}{2}\right)^2 + 2\pi\left(\frac{d}{2}\right)h.$$

Thus

$$A_P = \pi \left(\frac{x^2}{2} + xy \right), \quad A_Q = \pi \left(\frac{y^2}{2} + xy \right).$$

Set $x = ky$. Then

$$\frac{k^2 + 2k}{1 + 2k} = \frac{8}{5}.$$

So $5k^2 - 6k - 8 = 0$, giving the positive root $k = 2$. Hence $x = 2y$.

Now

$$V_P - V_Q = \frac{\pi}{4}(x^2y - y^2x) = \frac{\pi}{4}xy(x - y) = 32\pi.$$

Substitute $x = 2y$:

$$\frac{\pi}{4}(2y)(y)(y) = 32\pi \Rightarrow \frac{y^3}{2} = 32 \Rightarrow y^3 = 64 \Rightarrow y = 4.$$

Hence $x = 8$, so $x + y = \boxed{12}$.

M1-08

Ratio change after equal spending

Answer **D**

QUESTION 8

Pat and Alex have a combined total of £ 84. The ratio of Pat's money to Alex's money is 5 : 2.

They each spend the same amount on sweets.

After this, Pat receives £ 6 and Alex receives £ 1. The ratio of their new amounts is then 3 : 1.

How much did each of them spend?

A £ 3

B £ 4

C £ 4.25

D £ 4.50

E £ 6

WORKED SOLUTION

Initially the amounts are 60 and 24.

If each spends s , then afterwards they have $60 - s$ and $24 - s$. After receiving the extra money, they have $66 - s$ and $25 - s$.

So

$$66 - s = 3(25 - s) \Rightarrow 66 - s = 75 - 3s \Rightarrow 2s = 9 \Rightarrow s = 4.5.$$

Each spent £ 4.50.

M1-09

Coordinate geometry of a tangent

Answer A

QUESTION 9

The circle $x^2 + y^2 = 25$ has centre at the origin.

A tangent to the circle passes through the point $(1, 7)$ and has positive gradient.

What is the equation of the tangent?

A $3x - 4y + 25 = 0$

B $4x - 3y + 25 = 0$

C $3x + 4y - 25 = 0$

D $4x + 3y - 25 = 0$

E

$$y = \frac{4}{3}x + \frac{17}{3}$$

WORKED SOLUTION

Let the tangent be $y = mx + c$. Since it passes through $(1, 7)$, we have $c = 7 - m$.

The distance from the origin to the tangent must equal the radius 5:

$$\frac{|c|}{\sqrt{m^2 + 1}} = 5.$$

So

$$(7 - m)^2 = 25(m^2 + 1) \Rightarrow 12m^2 + 7m - 12 = 0 = (4m - 3)(3m + 4).$$

The positive gradient is $m = \frac{3}{4}$. Then $c = 7 - \frac{3}{4} = \frac{25}{4}$.

Hence

$$y = \frac{3}{4}x + \frac{25}{4} \Leftrightarrow \boxed{3x - 4y + 25 = 0}.$$

M1-10

Conditional probability with non-standard dice

Answer **E****QUESTION 10**

Two fair dice X and Y are available.

- The faces on X are labelled 1, 1, 2, 3, 4, 5.
- The faces on Y are labelled 2, 3, 3, 4, 5, 6.

One of the dice is chosen at random and rolled twice.

Given that the total shown is 7, what is the probability that the chosen die was X ?

A

$$\frac{1}{3}$$

B $\frac{3}{8}$

C $\frac{1}{2}$

D $\frac{3}{5}$

E $\frac{2}{5}$

WORKED SOLUTION

For die X, the ordered outcomes summing to 7 are (2, 5), (5, 2), (3, 4), (4, 3): 4 outcomes out of 36. So $P(7 | X) = \frac{1}{9}$.

For die Y, 3 appears twice, so the ordered outcomes summing to 7 are 6 in total. Hence $P(7 | Y) = \frac{1}{6}$.

With equal prior probability of choosing each die,

$$P(X | 7) = \frac{\frac{1}{2} \cdot \frac{1}{9}}{\frac{1}{2} \cdot \frac{1}{9} + \frac{1}{2} \cdot \frac{1}{6}} = \frac{\frac{1}{9}}{\frac{1}{9} + \frac{1}{6}} = \frac{2}{5}.$$

M1-11

Exponential equation via substitution

Answer **C**

QUESTION 11

The equation $2^x + 2^{2-x} = 5$ has two real solutions.

What is the larger possible value of x?

A 0

B 1

C 2

D 3

E 4

WORKED SOLUTION

Let $t = 2^x$. Then $2^{2-x} = \frac{4}{t}$, so

$$t + \frac{4}{t} = 5 \Rightarrow t^2 - 5t + 4 = 0 = (t - 1)(t - 4).$$

Thus $t = 1$ or $t = 4$, so $x = 0$ or $x = 2$. The larger value is 2.

M1-12

Rearranging to make a variable the subject

Answer B

QUESTION 12

Suppose x and y satisfy

$$x = y + \frac{4}{y},$$

where $y > 0$ and $x > 4$.

Which of the following is the larger possible value of y in terms of x ?

A $\frac{x - \sqrt{x^2 - 16}}{2}$

B $\frac{x + \sqrt{x^2 - 16}}{2}$

C $\frac{x + \sqrt{x^2 + 16}}{2}$

D $\frac{2}{x + \sqrt{x^2 - 16}}$

E $\frac{4}{x + \sqrt{x^2 - 16}}$

WORKED SOLUTION

Multiply by y :

$$xy = y^2 + 4 \Rightarrow y^2 - xy + 4 = 0.$$

Using the quadratic formula,

$$y = \frac{x \pm \sqrt{x^2 - 16}}{2}.$$

The larger possible value uses the positive sign, so

$$y = \frac{x + \sqrt{x^2 - 16}}{2}.$$

M1-13

Surface area comparison of a cylinder and a cube

Answer D

QUESTION 13

A solid cylinder has radius r cm and height h cm.

A cube has side length $\sqrt{\pi} r$ cm.

The total surface area of the cylinder is equal to four times the total surface area of the cube.

What is $\frac{h}{r}$?

A 7

B 9

C 10

D 11

E 12

WORKED SOLUTION

The cylinder's total surface area is $2\pi r^2 + 2\pi rh = 2\pi r(r + h)$.

The cube's total surface area is $6(\sqrt{\pi}r)^2 = 6\pi r^2$.

Given the cylinder equals four times the cube,

$$2\pi r(r + h) = 24\pi r^2.$$

Cancel $2\pi r$:

$$r + h = 12r \Rightarrow h = 11r.$$

Therefore $\frac{h}{r} = \boxed{11}$.

M1-14

Mean and median with an ordered list

Answer A

QUESTION 14

Consider the list

$$1, 2, x, x + 1, 2x - 1, 15,$$

where $2 < x < 8$.

The mean of the six numbers is one more than the median.

What is the value of x ?

A $\frac{9}{2}$

B 4

C 5

D $\frac{11}{2}$

E 6

WORKED SOLUTION

Since $x > 2$, the list is already ordered. So the median is

$$\frac{x + (x + 1)}{2} = x + \frac{1}{2}.$$

The mean is one more than this, so the mean is $x + \frac{3}{2}$.

The mean is also

$$\frac{1 + 2 + x + (x + 1) + (2x - 1) + 15}{6} = \frac{4x + 18}{6}.$$

Therefore

$$\frac{4x + 18}{6} = x + \frac{3}{2} \Rightarrow 4x + 18 = 6x + 9 \Rightarrow 2x = 9 \Rightarrow \boxed{x = \frac{9}{2}}.$$

QUESTION 15

A cyclist rides from A to B uphill at 12 km h^{-1} , immediately turns round, and returns from B to A along the same route at 18 km h^{-1} .

She spends exactly 6 minutes stationary at B.

The total time for the complete trip is 46 minutes.

What is the distance from A to B?

A 3.6 km

B 4.0 km

C 4.2 km

D 4.5 km

E 4.8 km

WORKED SOLUTION

Excluding the 6-minute stop, the travel time is 40 minutes = $\frac{2}{3}$ hours.

If the one-way distance is d km, then

$$\frac{d}{12} + \frac{d}{18} = \frac{2}{3}.$$

Since $\frac{1}{12} + \frac{1}{18} = \frac{5}{36},$

$$d \cdot \frac{5}{36} = \frac{2}{3} \Rightarrow d = \frac{2}{3} \cdot \frac{36}{5} = \frac{24}{5} = 4.8.$$

So the distance is 4.8 km.

M1-16

Exact trigonometric values

Answer **B**

QUESTION 16

Given that

$$y = \frac{2 \sin 15^\circ}{\cos 30^\circ},$$

what is the value of y^2 ?

A $\frac{4 - 2\sqrt{3}}{3}$

B $\frac{8 - 4\sqrt{3}}{3}$

C $\frac{8 + 4\sqrt{3}}{3}$

D $2 - \sqrt{3}$

E $4 - 2\sqrt{3}$

WORKED SOLUTION

Use

$$\sin 15^\circ = \sin(45^\circ - 30^\circ) = \frac{\sqrt{6} - \sqrt{2}}{4}, \quad \cos 30^\circ = \frac{\sqrt{3}}{2}.$$

Hence

$$y = \frac{2 \cdot \frac{\sqrt{6} - \sqrt{2}}{4}}{\frac{\sqrt{3}}{2}} = \frac{\sqrt{6} - \sqrt{2}}{\sqrt{3}}.$$

Therefore

$$y^2 = \frac{(\sqrt{6} - \sqrt{2})^2}{3} = \frac{6 + 2 - 2\sqrt{12}}{3} = \frac{8 - 4\sqrt{3}}{3}.$$

M1-17

Inverse proportion in standard form

Answer C

QUESTION 17

The variable t is inversely proportional to the square of w .

When $w = 2 \times 10^3$, the value of t is 4.5×10^{-5} .

What is the value of w when $t = 8 \times 10^{-6}$?

A 7.5×10^3

B $3\sqrt{10} \times 10^2$

C $1.5\sqrt{10} \times 10^3$

D $2\sqrt{10} \times 10^3$

E 4.5×10^3

WORKED SOLUTION

Since $t = \frac{k}{w^2}$, use the first condition:

$$k = (4.5 \times 10^{-5})(2 \times 10^3)^2 = (4.5 \times 10^{-5})(4 \times 10^6) = 180.$$

Now solve

$$8 \times 10^{-6} = \frac{180}{w^2} \Rightarrow w^2 = 2.25 \times 10^7.$$

So

$$w = \sqrt{2.25 \times 10^7} = 1.5\sqrt{10} \times 10^3.$$

M1-18

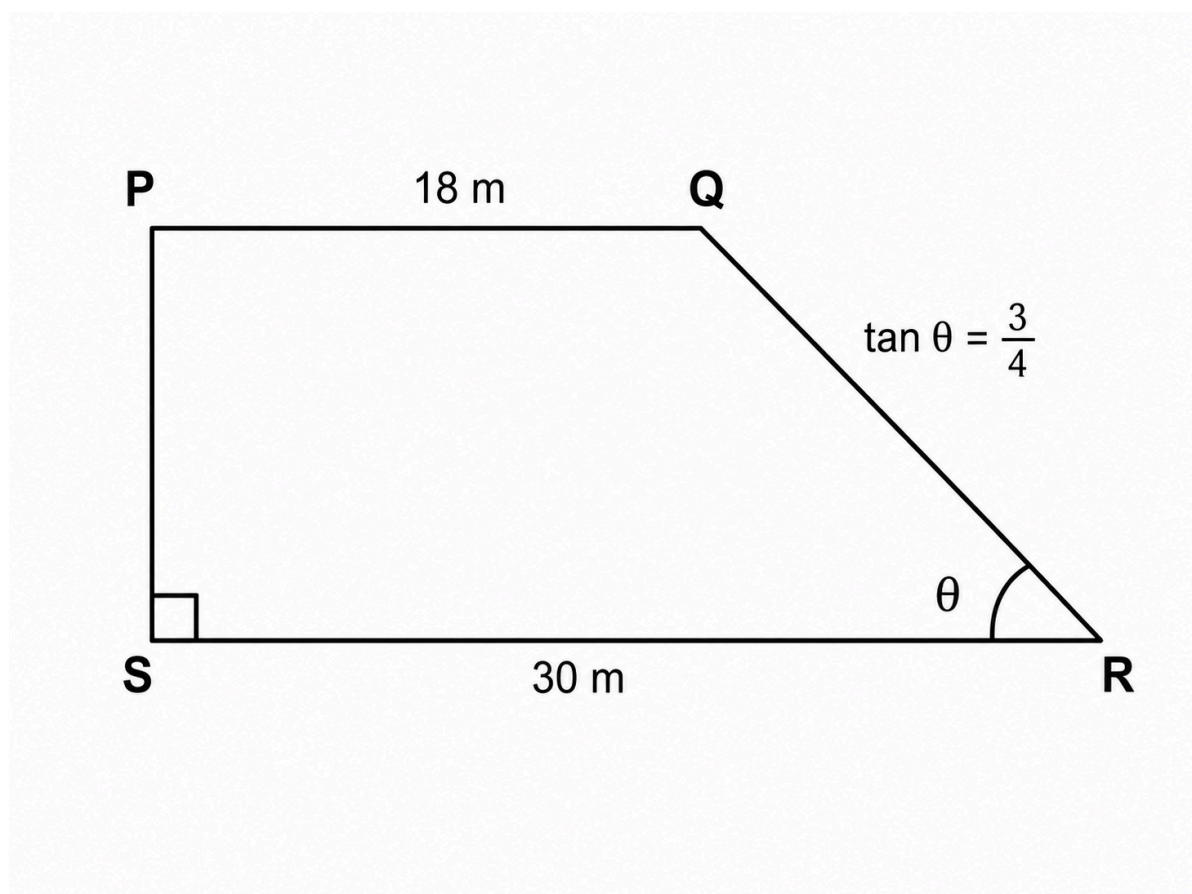
Trapezium area using trigonometry

Answer D

QUESTION 18

PQRS is a trapezium in which $PQ \parallel SR$ and $PS \perp SR$.

Also, $PQ = 18 \text{ m}$, $SR = 30 \text{ m}$, and $\tan \angle QRS = \frac{3}{4}$.



What is the area of the trapezium?

A 180 m^2

B 192 m^2

C 204 m^2

D 216 m^2

E 240 m^2

WORKED SOLUTION

The difference between the parallel sides is $30 - 18 = 12$, which is the horizontal run of the sloping side.

If the height is h , then

$$\tan \angle QRS = \frac{h}{12} = \frac{3}{4} \Rightarrow h = 9.$$

So the area is

$$\frac{1}{2}(18 + 30)(9) = 24 \cdot 9 = \boxed{216 \text{ m}^2}.$$

M1-19

Quadratic roots and discriminant reasoning

Answer A

QUESTION 19

The equation $x^2 + bx + 8 = 0$ has two real roots whose positive difference is 2. The sum of the roots is negative.

What is the value of b ?

A 6

B 5

C 4

D -6

E -4

WORKED SOLUTION

If the roots are r and s , then $(r - s)^2 = (r + s)^2 - 4rs$.

Here $r + s = -b$ and $rs = 8$. Since the difference is 2,

$$4 = b^2 - 32 \Rightarrow b^2 = 36.$$

So $b = \pm 6$. The sum of the roots is negative, so $-b < 0 \Rightarrow b > 0$. Hence

$$\boxed{b = 6}.$$

M1-20

Algebraic geometry with area change

Answer **E**

QUESTION 20

A rectangle has length $(2x - 1)$ cm and width $(x + 1)$ cm.

A larger rectangle is made by adding 3 cm to both the length and the width.

The larger rectangle has twice the area of the original rectangle.

What is the perimeter of the original rectangle?

A 24 cm

B 26 cm

C 28 cm

D 29 cm

E

30 cm

WORKED SOLUTION

Original area is $(2x - 1)(x + 1)$. New dimensions are $2x + 2$ and $x + 4$, so the new area is $(2x + 2)(x + 4)$.

Given new area is twice old area,

$$(2x + 2)(x + 4) = 2(2x - 1)(x + 1).$$

This simplifies to

$$x^2 - 4x - 5 = 0 = (x - 5)(x + 1).$$

So $x = 5$. The original dimensions are 9 and 6, so the perimeter is

$$2(9 + 6) = \boxed{30 \text{ cm}}.$$

M1-21

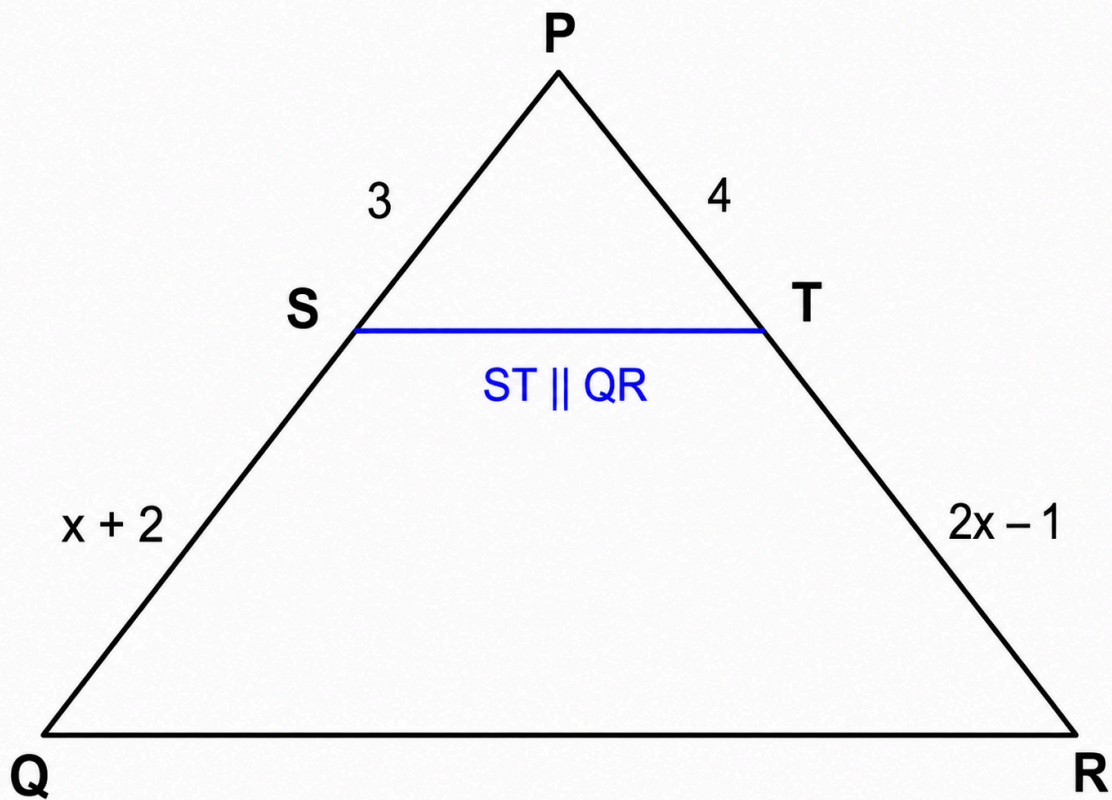
Similarity and area scaling

Answer C

QUESTION 21

In triangle PQR, point S lies on PQ and point T lies on PR. The line ST is parallel to QR.

The lengths are labelled as follows: PS = 3 cm, SQ = $(x + 2)$ cm, PT = 4 cm, TR = $(2x - 1)$ cm.



Given that the area of triangle PST is 12 cm^2 , what is the area of trapezium $STQR$?

A 96 cm^2

B 120 cm^2

C 135 cm^2

D 144 cm^2

E 147 cm^2

WORKED SOLUTION

Since $ST \parallel QR$, triangles PST and PQR are similar, so

$$\frac{PS}{PQ} = \frac{PT}{PR}.$$

Now $PQ = 3 + (x + 2) = x + 5$ and $PR = 4 + (2x - 1) = 2x + 3$.

Therefore

$$\frac{3}{x + 5} = \frac{4}{2x + 3}.$$

So $6x + 9 = 4x + 20$, giving $x = \frac{11}{2}$.

Then

$$\frac{PS}{PQ} = \frac{3}{21/2} = \frac{2}{7}.$$

Hence the area scale factor is $\left(\frac{2}{7}\right)^2 = \frac{4}{49}$.

So if area $PST = 12$, then area $PQR = 12 \cdot \frac{49}{4} = 147$.

Thus the area of trapezium $STQR$ is $147 - 12 = \boxed{135 \text{ cm}^2}$.

M1-22

Probability with two colours

Answer **B**

QUESTION 22

Charlie has a bowl containing red sweets and green sweets only. There are 9 more red sweets than green sweets.

Two sweets are chosen at random without replacement.

The probability that the two sweets are the same colour is $\frac{4}{7}$.

How many sweets are there in the bowl altogether?

A 18

B 21

C 24

D 27

E 30

WORKED SOLUTION

Let the number of green sweets be g . Then the number of red sweets is $g + 9$, so total sweets are $2g + 9$.

The probability condition is

$$\frac{\binom{g+9}{2} + \binom{g}{2}}{\binom{2g+9}{2}} = \frac{4}{7}.$$

This simplifies to

$$(g - 6)(g + 18) = 0.$$

So $g = 6$. Then red sweets = 15, giving total sweets = 21.

M1-23

Successive translations and vector equations

Answer **D**

QUESTION 23

The point $(-1, 5)$ is translated by two successive vectors.

The first translation is by $\begin{pmatrix} 3p \\ 2 - q \end{pmatrix}$, and the second is by $\begin{pmatrix} q + 1 \\ p - 3 \end{pmatrix}$.

The final image is $(8, 4)$.

What is the value of $p^2 + q^2$?

A 4

B 5

C 6

D 8

E 10

WORKED SOLUTION

The total translation vector is

$$\begin{pmatrix} 3p + q + 1 \\ p - q - 1 \end{pmatrix}.$$

Since $(-1, 5) \rightarrow (8, 4)$, the net translation is $\begin{pmatrix} 9 \\ -1 \end{pmatrix}$.

So

$$3p + q = 8, \quad p - q = 0.$$

Thus $p = q$, so $4p = 8 \Rightarrow p = 2$, hence $q = 2$.

Therefore $p^2 + q^2 = 4 + 4 = \boxed{8}$.

M1-24

Quadratic graphs and minimum values

Answer **A**

QUESTION 24

Consider the family of graphs

$$y = ax^2 + 2(a - 1)x + a + 3.$$

What is the complete range of values of a for which the minimum point of the graph lies above the x -axis?

A $a > \frac{1}{5}$

B $a > 0$

C $a \geq \frac{1}{5}$

D $a < 0$

E $0 < a < \frac{1}{5}$

WORKED SOLUTION

For there to be a minimum, we need $a > 0$.

For $y = ax^2 + bx + c$, the minimum value is $c - \frac{b^2}{4a}$.

Here $b = 2(a - 1)$ and $c = a + 3$, so the minimum value is

$$a + 3 - \frac{(2(a - 1))^2}{4a} = a + 3 - \frac{(a - 1)^2}{a} = 5 - \frac{1}{a}.$$

We need this to be positive:

$$5 - \frac{1}{a} > 0.$$

Since $a > 0$, this gives $5a > 1 \Rightarrow a > \frac{1}{5}$.

M1-25

Counting constrained selections

Answer **E**

QUESTION 25

An online company sells storage containers. The following items are available:

Capacity of container	Number available
2 litres	2
3 litres	3

Capacity of container	Number available
5 litres	4

A customer chooses a selection of containers with total capacity 23 litres.

Different selections are distinguished only by how many containers of each size are chosen.

If all possible selections are equally likely, what is the probability that the chosen selection contains at least one 2-litre container?

A $\frac{1}{3}$

B $\frac{1}{2}$

C $\frac{3}{5}$

D $\frac{3}{4}$

E $\frac{2}{3}$

WORKED SOLUTION

Let a, b, c be the numbers of 2-, 3- and 5-litre containers. Then

$$2a + 3b + 5c = 23,$$

with $0 \leq a \leq 2, 0 \leq b \leq 3, 0 \leq c \leq 4$.

The valid selections are

- (0, 1, 4)
- (1, 2, 3)
- (2, 3, 2)

Two of the three contain at least one 2-litre container, so the probability is $\frac{2}{3}$.

M1-26

Similarity and three-dimensional scaling

Answer C

QUESTION 26

The space diagonal of a cuboid is $7\sqrt{13}$ cm.

A similar cuboid has volume 8 times the volume of the original cuboid.

What is the space diagonal of the larger cuboid?

A $7\sqrt{26}$ cm

B $14\sqrt{7}$ cm

C $14\sqrt{13}$ cm

D $21\sqrt{13}$ cm

E $28\sqrt{13}$ cm

WORKED SOLUTION

A volume scale factor of 8 means a linear scale factor of $\sqrt[3]{8} = 2$.

So the new diagonal is

$$2 \cdot 7\sqrt{13} = \boxed{14\sqrt{13} \text{ cm}}.$$

QUESTION 27

Alex, Cameron and Sam are running a 400 m race. They each run at a different constant speed.

When Alex finishes, Cameron has run 300 m and Sam has run 240 m.

Alex beats Sam by $33\frac{1}{3}$ seconds.

What is Cameron's speed?

A 5 m s^{-1}

B 6 m s^{-1}

C 6.4 m s^{-1}

D 7 m s^{-1}

E 8 m s^{-1}

WORKED SOLUTION

Let Alex's speed be A , Cameron's speed be C , and Sam's speed be S .

When Alex finishes, the elapsed time is $\frac{400}{A}$. In that time, Cameron runs 300 m, so

$$C \cdot \frac{400}{A} = 300 \Rightarrow \frac{C}{A} = \frac{3}{4}.$$

Similarly, Sam runs 240 m, so

$$S \cdot \frac{400}{A} = 240 \Rightarrow \frac{S}{A} = \frac{3}{5}.$$

$$\text{Thus } S = \frac{3}{5}A.$$

Alex beats Sam by $\frac{100}{3}$ s, so

$$\frac{400}{S} - \frac{400}{A} = \frac{100}{3}.$$

Substitute $S = \frac{3}{5}A$:

$$\frac{400}{3A/5} - \frac{400}{A} = \frac{100}{3} \Rightarrow \frac{2000}{3A} - \frac{400}{A} = \frac{100}{3} \Rightarrow \frac{800}{3A} = \frac{100}{3}.$$

So $A = 8 \text{ m s}^{-1}$. Hence

$$C = \frac{3}{4}A = 6 \text{ m s}^{-1}.$$

Therefore Cameron's speed is 6 m s^{-1} .

Mathematics 2

27 QUESTIONS • COMPLETE SOLUTIONS

Engineering Mock Test 5 · preserved Mathematics 2 module. Question wording, option order, answer and solution method are unchanged.

M2-01

Polynomials

Answer **D****QUESTION 1**

The polynomial $P(x) = x^4 + ax^3 + bx^2 - 44x + 24$ has $(x - 1)$ and $(x - 2)$ as factors. What is the value of $a^2 + b$?

A 49**B** 61**C** 68**D** 75**E** 82**WORKED SOLUTION**

Since $x = 1$ and $x = 2$ are roots, $P(1)=0$ and $P(2)=0$.

From $P(1)=0$: $1 + a + b - 44 + 24 = 0$, so $a + b = 19$.

From $P(2)=0$: $16 + 8a + 4b - 88 + 24 = 0$, so $2a + b = 12$.

Subtracting gives $a = -7$, and then $b = 26$.

Hence $a^2 + b = 49 + 26 = \mathbf{75}$.

M2-02

Calculus / stationary points

Answer **B****QUESTION 2**

The curve $y = x^3 + px^2 + qx + 10$ has a local maximum at $x = -1$ and a local minimum at $x = 3$. What is the sum of the y-coordinates of these two stationary points?

A -8**B** -2**C** 2**D** 6**E** 12**WORKED SOLUTION**

Differentiate: $y' = 3x^2 + 2px + q$. Since the stationary points are at $x = -1$ and $x = 3$, we have $y' = 3(x+1)(x-3) = 3x^2 - 6x - 9$.

So $2p = -6$ and $q = -9$, hence $p = -3$.

The curve is $y = x^3 - 3x^2 - 9x + 10$.

At $x = -1$, $y = -1 - 3 + 9 + 10 = 15$.

At $x = 3$, $y = 27 - 27 - 27 + 10 = -17$.

The sum is $15 + (-17) = -2$.

M2-03

Coordinate geometry

Answer **E****QUESTION 3**

The line $y = mx + 3$ is reflected in the line $y = x$. The reflected line has y-intercept $-\frac{2}{3}$. What is the value of m ?

A $\frac{3}{2}$

B 2

C $\frac{5}{2}$

D 3

E $\frac{9}{2}$

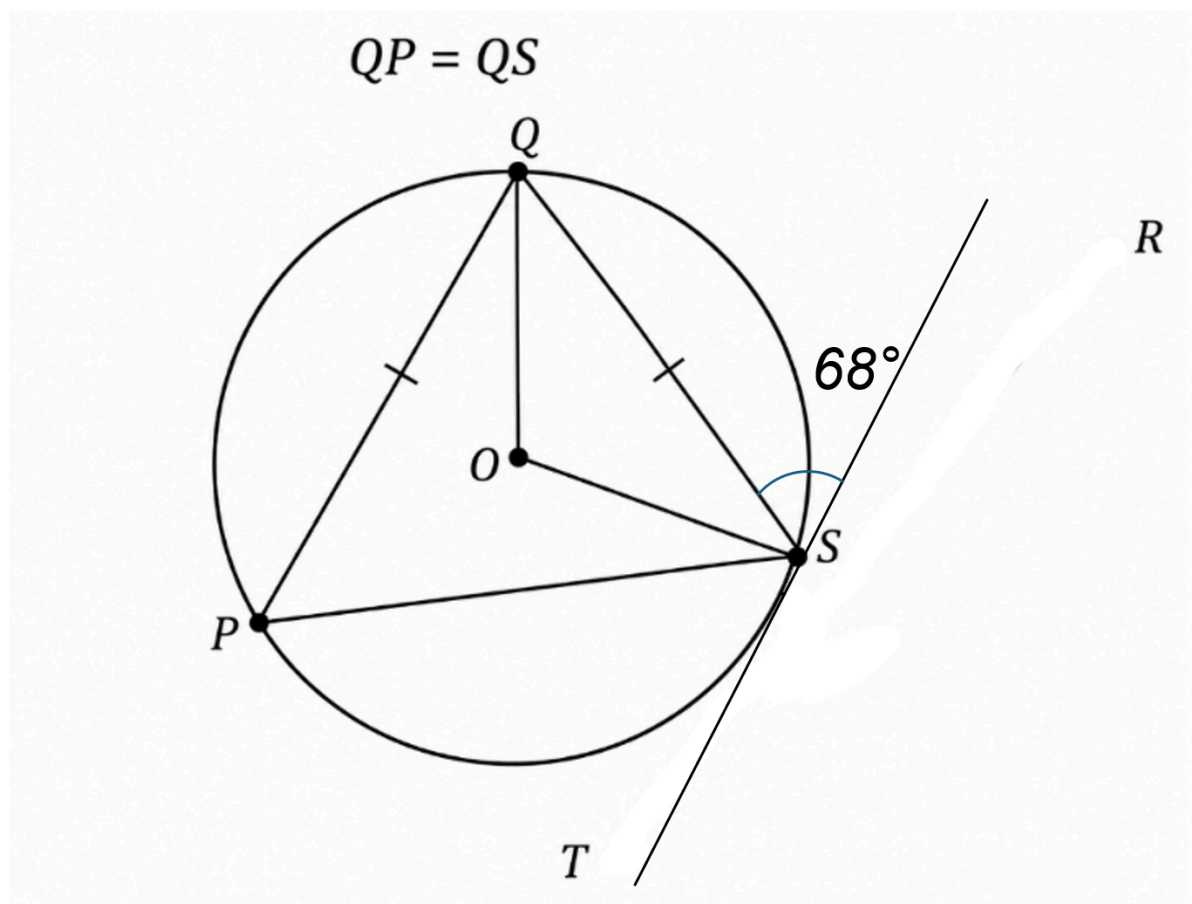
WORKED SOLUTION

Reflecting in $y = x$ swaps x and y . So the reflected line is $x = my + 3$, which rearranges to $y = (x - 3)/m = (1/m)x - 3/m$.

The y -intercept of the reflected line is therefore $-3/m$.

Given $-3/m = -2/3$, we get $3/m = 2/3$, so $9 = 2m$ and $m = \mathbf{9/2}$.

QUESTION 4



In the diagram, O is the centre of the circle and RT is a tangent at S . Points P and Q lie on the circumference, $QP = QS$, and the marked angle is $\angle QSR = 68^\circ$. What is the size of $\angle POS$?

A 44°

B 68°

C 88°

D 112°

E 136°

WORKED SOLUTION

By the tangent–chord theorem applied to chord QS,

$$\angle QPS = \angle QSR = 68^\circ.$$

Since $QP = QS$, triangle PQS is isosceles with base PS. Hence

$$\angle PSQ = 68^\circ, \quad \angle PQS = 180^\circ - 68^\circ - 68^\circ = 44^\circ.$$

Chord PS subtends angle $\angle PQS$ at the circumference and angle $\angle POS$ at the centre. Therefore

$$\angle POS = 2\angle PQS = 2(44^\circ) = \boxed{88^\circ}.$$

M2-05

Integration / area

Answer **B**

QUESTION 5

What is the total area enclosed between the curve $y = x^2 - 4x + 3$ and the line $y = -x + 1$, for $0 \leq x \leq 3$?

A $\frac{7}{6}$

B $\frac{11}{6}$

C $\frac{13}{6}$

D $\frac{17}{6}$

E $\frac{23}{6}$

WORKED SOLUTION

Subtract the line from the curve: $(x^2 - 4x + 3) - (-x + 1) = x^2 - 3x + 2 = (x-1)(x-2)$.

This is positive on $[0, 1]$, negative on $[1, 2]$, and positive on $[2, 3]$.

So total area = $\int_0^1 (x^2 - 3x + 2) dx - \int_1^2 (x^2 - 3x + 2) dx + \int_2^3 (x^2 - 3x + 2) dx$.

An antiderivative is $x^3/3 - 3x^2/2 + 2x$.

Evaluating gives $5/6 + 1/6 + 5/6 = 11/6$.

M2-06

Surds

Answer C

QUESTION 6

Simplify $\sqrt{(18 + 8\sqrt{5})} - \sqrt{(18 - 8\sqrt{5})}$. Write the answer in the form $k\sqrt{2}$. What is k ?

A 2

B 3

C 4

D 5

E 6

WORKED SOLUTION

Write $18 + 8\sqrt{5}$ as $a + b + 2\sqrt{ab}$ with $a + b = 18$ and $ab = 80$. The numbers are 10 and 8.

So $\sqrt{(18 + 8\sqrt{5})} = \sqrt{10 + \sqrt{8}} = \sqrt{10 + 2\sqrt{2}}$.

Similarly, $\sqrt{(18 - 8\sqrt{5})} = \sqrt{10 - 2\sqrt{2}}$.

Subtracting gives $(\sqrt{10 + 2\sqrt{2}}) - (\sqrt{10 - 2\sqrt{2}}) = 4\sqrt{2}$.

Hence $k = 4$.

M2-07

Tangents

Answer B

QUESTION 7

The line $y = 2x + k$ is a tangent to the curve $y = 3x^2 - 10x + 7$. What is the value of k ?

A -7**B** -5**C** -3**D** 1**E** 5**WORKED SOLUTION**

At the point of tangency the line and curve have one repeated intersection.

Set them equal: $3x^2 - 10x + 7 = 2x + k$, so $3x^2 - 12x + (7 - k) = 0$.

For tangency the discriminant is 0:

$$(-12)^2 - 4 \cdot 3 \cdot (7 - k) = 0.$$

$$144 - 12(7 - k) = 0 \Rightarrow 144 - 84 + 12k = 0 \Rightarrow 60 + 12k = 0.$$

So $k = -5$.

M2-08

Differentiation

Answer B

QUESTION 8

Given $y = (3\sqrt{x} - 2/\sqrt{x})^2$, find dy/dx when $x = \frac{1}{2}$.

A -9

B -7

C -5

D 7

E 9

WORKED SOLUTION

First simplify: $y = 9x - 12 + 4/x$.

Differentiate: $dy/dx = 9 - 4/x^2$.

At $x = 1/2$, $x^2 = 1/4$, so $4/x^2 = 16$.

Therefore $dy/dx = 9 - 16 = -7$.

M2-09

Integration

Answer **C**

QUESTION 9

Find the value of $\int_1^4 (3x^2 + 1)/(x\sqrt{x}) \, dx$.

A 12

B 14

C 15

D 16

E 18

WORKED SOLUTION

Rewrite the integrand: $(3x^2 + 1)/(x\sqrt{x}) = 3x^{1/2} + x^{-3/2}$.

So $\int (3x^{1/2} + x^{-3/2}) dx = 2x^{3/2} - 2x^{-1/2}$.

Evaluate from 1 to 4:

$$[2x^{3/2} - 2x^{-1/2}]_1^4 = (2 \cdot 8 - 2 \cdot 1/2) - (2 \cdot 1 - 2 \cdot 1) = 15 - 0 = 15.$$

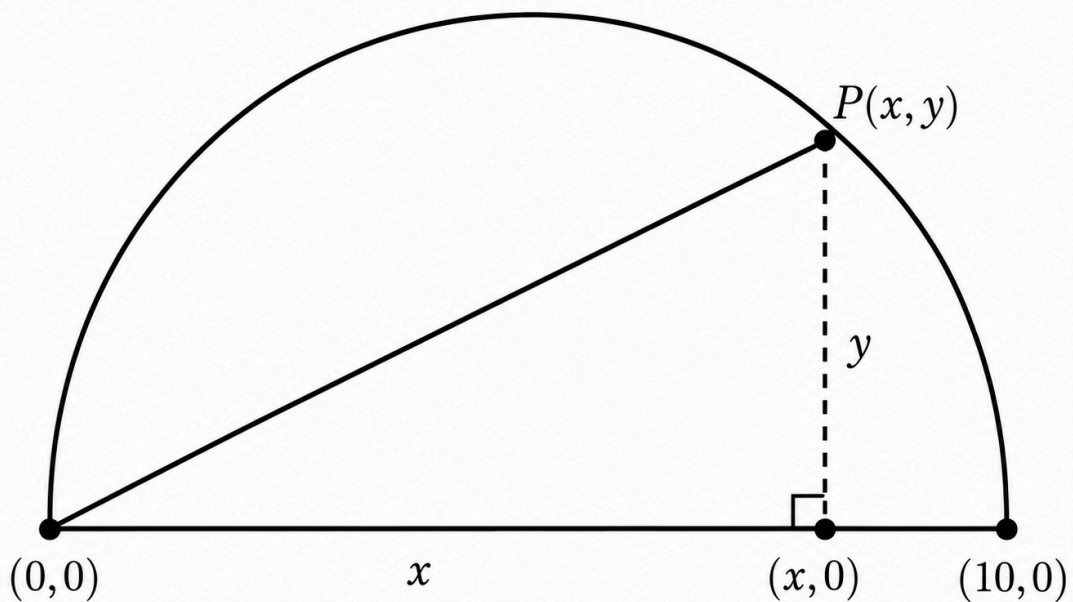
M2-10

Coordinate geometry / optimisation

Answer **B**

QUESTION 10

P lies on the semicircle of radius 5



A point $P(x,y)$ lies on the semicircle with diameter from $(0,0)$ to $(10,0)$. A right triangle has vertices at $(0,0)$, $(x,0)$, and $P(x,y)$. What is the maximum possible area of the triangle?

A $25\sqrt{3/8}$

B $75\sqrt{3/8}$

C $25\sqrt[3]{4}$

D $75\sqrt[3]{4}$

E $125\sqrt[3]{8}$

WORKED SOLUTION

The semicircle has centre $(5,0)$ and radius 5, so $y = \sqrt{(25 - (x-5)^2)} = \sqrt{(10x - x^2)}$.

The triangle area is $A = \frac{1}{2}xy = \frac{1}{2}x\sqrt{(10x - x^2)}$.

To maximise A , maximise $A^2 = \frac{1}{4}x^2(10x - x^2) = \frac{1}{4}(10x^3 - x^4)$.

Differentiate: $d(A^2)/dx = \frac{1}{4}(30x^2 - 4x^3) = x^2(15 - 2x)/2$.

So the maximum occurs at $x = 15/2$.

Then $y = \sqrt{(10 \cdot 15/2 - (15/2)^2)} = \sqrt{(75/4)} = 5\sqrt{3}/2$.

Hence $A = \frac{1}{2} \cdot 15/2 \cdot 5\sqrt{3}/2 = 75\sqrt{3}/8$.

M2-11

Inequalities

Answer **B**

QUESTION 11

What is the complete set of real values of x for which $x^2(x^2 + 4) < 21$?

A $-2 < x < 2$

B $-\sqrt{3} < x < \sqrt{3}$

C $x < -\sqrt{3}$ or $x > \sqrt{3}$

D $-\sqrt{7} < x < \sqrt{7}$

E $-1 < x < 1$

WORKED SOLUTION

Let $u = x^2$, so $u \geq 0$. Then the inequality becomes $u(u+4) < 21$, i.e. $u^2 + 4u - 21 < 0$.

Factor: $(u+7)(u-3) < 0$, so $-7 < u < 3$.

Since $u = x^2 \geq 0$, we get $0 \leq u < 3$.

Thus $x^2 < 3$, so $-\sqrt{3} < x < \sqrt{3}$.

M2-12

Similar solids

Answer B

QUESTION 12

Four mathematically similar solids W, X, Y and Z satisfy the following:

- the ratio of lengths W:X is 1:2
- the ratio of surface areas X:Y is 4:9
- the ratio of volumes Y:Z is 8:27

What is the ratio of the volumes of W:Z?

A 1:81**B** 8:729**C** 4:243**D** 8:243**E** 16:729**WORKED SOLUTION**

Length ratio W:X = 1:2.

Surface area ratio X:Y = 4:9, so length ratio X:Y = 2:3.

Volume ratio Y:Z = 8:27, so length ratio Y:Z = 2:3.

Hence the overall length ratio $W:Z = 1:2 \times 2:3 \times 2:3 = 2:9$.

Volumes scale as the cube of length, so $W:Z$ volume ratio = $2^3:9^3 = 8:729$.

M2-13

Logarithms

Answer C

QUESTION 13

What is the mean of $\log_2 64$, $\log_4 64$, and $\log_8 64$?

A 3

B $10/3$

C $11/3$

D 4

E $13/3$

WORKED SOLUTION

$\log_2 64 = 6$, $\log_4 64 = 3$, and $\log_8 64 = 2$.

So the mean is $(6 + 3 + 2)/3 = 11/3$.

M2-14

Trigonometric comparison

Answer B

QUESTION 14

Which of the following is largest in value? (All angles are in radians.)

A $\sin 1$

B $\cos 0.5$

C $\tan 0.5$

D $\sin 0.5$

E $\cos 1$

WORKED SOLUTION

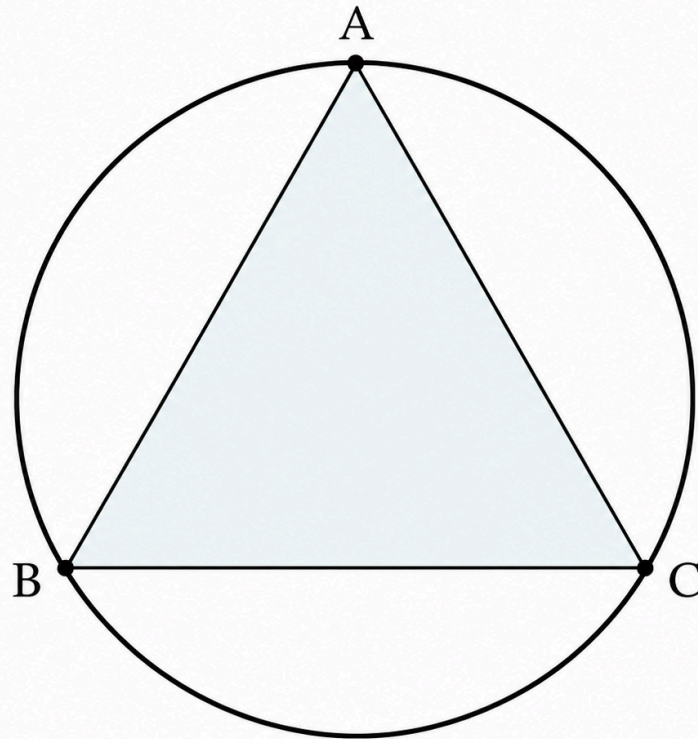
On $0 < \theta < \pi/2$, $\sin \theta$ and $\tan \theta$ increase, while $\cos \theta$ decreases.

Now $\sin 1 = \cos(\pi/2 - 1) = \cos(0.5708\dots)$. Since $0.5 < 0.5708\dots$, and cosine decreases on this interval, $\cos 0.5 > \cos 0.5708\dots = \sin 1$.

Also $\tan 0.5 < 1$ and in fact $\tan 0.5 < \cos 0.5$ numerically, while $\sin 0.5$ and $\cos 1$ are smaller still.

So the largest value is **$\cos 0.5$** .

QUESTION 15



$$AB = BC = CA = 6$$

An equilateral triangle of side 6 cm is inscribed in a circle. What is the area inside the circle but outside the triangle?

A $12\pi - 6\sqrt{3}$

B $12\pi - 9\sqrt{3}$

C $6\pi - 9\sqrt{3}$

D $4\pi - 9\sqrt{3}$

E $12\pi - 18\sqrt{3}$

WORKED SOLUTION

For an equilateral triangle of side s , the circumradius is $R = s/\sqrt{3}$.

So here $R = 6/\sqrt{3} = 2\sqrt{3}$.

Circle area $= \pi R^2 = \pi(2\sqrt{3})^2 = 12\pi$.

Triangle area $= (\sqrt{3}/4)s^2 = (\sqrt{3}/4) \cdot 36 = 9\sqrt{3}$.

Required area $= 12\pi - 9\sqrt{3}$, so the answer is **$12\pi - 9\sqrt{3}$** .

M2-16

Sequences and series

Answer C

QUESTION 16

An arithmetic sequence has first term a and common difference a . If S_n is the sum of the first n terms, which of the following is equal to $S_{2n} - 3S_n$?

A $an(n+1)/2$

B $an(2n-1)/2$

C $an(n-1)/2$

D $a(n-1)(n+1)$

E an^2

WORKED SOLUTION

The terms are $a, 2a, 3a, \dots, na$, so $S_n = a(1 + 2 + \dots + n) = an(n+1)/2$.

Then $S_{2n} = a \cdot 2n(2n+1)/2 = an(2n+1)$.

So $S_{2n} - 3S_n = an(2n+1) - 3[an(n+1)/2]$

$= [2an(2n+1) - 3an(n+1)]/2 = an(n-1)/2$.

Thus the correct expression is **$an(n-1)/2$** .

M2-17

Exponential equations

Answer C

QUESTION 17

The equation $2 \cdot 5^{2x} - 7 \cdot 5^x + 3 = 0$ has two real solutions. What is the sum of these two solutions?

A $\log_5 3$

B $\log_5(1/2)$

C $\log_5(3/2)$

D $\log_5 6$

E $\log_5(2/3)$

WORKED SOLUTION

Let $u = 5^x$. Then $2u^2 - 7u + 3 = 0$, which factors as $(2u-1)(u-3)=0$.

So $u = 1/2$ or $u = 3$.

Hence $x = \log_5(1/2)$ or $x = \log_5 3$.

The sum is $\log_5(1/2) + \log_5 3 = \log_5(3/2)$.

Therefore the answer is **$\log_5(3/2)$** .

M2-18

Coordinate geometry

Answer D

QUESTION 18

The line L with equation $y = mx + c$, where $m > 0$ and $c \geq 0$, passes through the point $(2,4)$. A line is drawn through $(2,4)$ perpendicular to L . The triangle enclosed between the two lines and the y -axis has area 5 square units. What is the larger possible value of m ?

A $\frac{1}{2}$

B 1

C $\frac{3}{2}$

D 2

E $\frac{5}{2}$

WORKED SOLUTION

Since L passes through $(2,4)$, its equation is $y = mx + (4 - 2m)$. So its y -intercept is $4 - 2m$.

The perpendicular line through $(2,4)$ has slope $-1/m$, so its equation is $y - 4 = -(1/m)(x - 2)$.

Its y -intercept is $4 + 2/m$.

The base of the triangle on the y -axis has length $(4 + 2/m) - (4 - 2m) = 2m + 2/m$.

The horizontal distance from $(2,4)$ to the y -axis is 2, so

Area = $\frac{1}{2} \times (2m + 2/m) \times 2 = 2(m + 1/m)$.

Set this equal to 5:

$$2(m + 1/m) = 5 \Rightarrow 2m^2 - 5m + 2 = 0 \Rightarrow (2m-1)(m-2)=0.$$

Thus $m = 1/2$ or 2, and the larger value is **2**.

M2-19

Algebraic structure

Answer **D**

QUESTION 19

How many distinct real solutions does the equation $(x^2 - 2x - 3)^2 = x^2 - 2x - 3$ have?

A 1

B 2

C 3

D 4

E 5

WORKED SOLUTION

Let $u = x^2 - 2x - 3$. Then the equation becomes $u^2 = u$, so $u(u-1)=0$.

Hence $u = 0$ or $u = 1$.

If $u = 0$, then $x^2 - 2x - 3 = 0 \Rightarrow (x-3)(x+1)=0 \Rightarrow x = 3$ or -1 .

If $u = 1$, then $x^2 - 2x - 4 = 0 \Rightarrow x = 1 \pm \sqrt{5}$.

These are four distinct real solutions, so the answer is **4**.

M2-20

Quadratics

Answer **C**

QUESTION 20

The curve $y = x^2 - x - 6$ intersects the x -axis at the points A and B, and has a minimum at the point C. A rectangle ABDE has two of its vertices at A and B, and C lies on the edge DE, between D and E. What is the area of the rectangle?

A 25

B $125/8$

C $125/4$

D 35

E 45

WORKED SOLUTION

The roots of $x^2 - x - 6 = 0$ are $x = 3$ and $x = -2$, so $AB = 5$.

The x-coordinate of the minimum is $x = -b/(2a) = 1/2$.

Then y at C is $(1/2)^2 - 1/2 - 6 = 1/4 - 1/2 - 6 = -25/4$.

Since C lies on the top edge DE of the rectangle while A and B lie on the x -axis, the height of the rectangle is $25/4$.

Area = base $\times$ height = $5 \times 25/4 = 125/4$.

M2-21

Trigonometric equations

Answer C

QUESTION 21

It is given that $9\cos x + \tan x \cdot \sin x = 6$, where $0^\circ < x < 90^\circ$. What are the possible values of $\tan x$?

A $\frac{1}{2}$ or $\frac{1}{4}$

B $\frac{1}{\sqrt{3}}$ or $\frac{1}{\sqrt{15}}$

C $\sqrt{3}$ or $\sqrt{15}$

D 2 or 4

E $\sqrt{3}$ or $2\sqrt{2}$

WORKED SOLUTION

Let $c = \cos x$. Then $\tan x \cdot \sin x = \sin^2 x / \cos x = (1 - c^2)/c$.

So the equation becomes $9c + (1 - c^2)/c = 6$.

Multiply by c : $9c^2 + 1 - c^2 = 6c$, so $8c^2 - 6c + 1 = 0$.

Factor: $(4c - 1)(2c - 1) = 0$, giving $c = 1/4$ or $c = 1/2$.

If $\cos x = 1/2$, then $\tan x = \sqrt{3}$.

If $\cos x = 1/4$, then $\sin x = \sqrt{1 - 1/16} = \sqrt{15}/4$, so $\tan x = (\sqrt{15}/4)/(1/4) = \sqrt{15}$.

Therefore the possible values are **$\sqrt{3}$ or $\sqrt{15}$** .

M2-22

Distance from point to line

Answer **B****QUESTION 22**

A straight line passes through the points $(0, 3a)$ and $(2a, 0)$, where a is a positive constant. What is the perpendicular distance of the point $P(a, 4a)$ from this line?

A $5a/\sqrt{13}$

B $5\sqrt{13} a/13$

C $2\sqrt{13} a/13$

D $4\sqrt{13} a/13$

E $3\sqrt{13} a/13$

WORKED SOLUTION

The line through $(0, 3a)$ and $(2a, 0)$ has intercept form $x/(2a) + y/(3a) = 1$, so $3x + 2y - 6a = 0$.

The perpendicular distance from $(a, 4a)$ to this line is

$$|3a + 2(4a) - 6a| / \sqrt{(3^2 + 2^2)} = |5a|/\sqrt{13} = 5a/\sqrt{13} = \mathbf{5\sqrt{13} a/13}.$$

M2-23

Intersections of curves

Answer **C****QUESTION 23**

At how many distinct points do the curves $y = (x - 3)(x^2 - 4x - 5)$ and $y = -x^2 + 6x - 9$ meet?

A 1

B 2

C 3

D 4

E 5

WORKED SOLUTION

Set the equations equal:

$$(x - 3)(x^2 - 4x - 5) = -x^2 + 6x - 9.$$

Bring everything to one side:

$$(x - 3)(x^2 - 4x - 5) + x^2 - 6x + 9 = 0.$$

Since $x^2 - 6x + 9 = (x-3)^2$, factor out $(x-3)$:

$$(x-3)[x^2 - 4x - 5 + (x-3)] = (x-3)(x^2 - 3x - 8) = 0.$$

So $x = 3$, or $x = (3 \pm \sqrt{41})/2$.

These are three distinct x -values, so the curves meet at **3** distinct points.

M2-24

Geometric progressions

Answer **C**

QUESTION 24

A geometric progression has first term $u_1 = a$ and common ratio r . The sum to infinity is $\frac{8}{5}$, and the sum to infinity of the even-numbered terms ($u_2 + u_4 + u_6 + \dots$) is $\frac{3}{5}$. What is the value of $a + r$?

A $\frac{21}{25}$

B $\frac{6}{5}$

C $\frac{31}{25}$

D $\frac{8}{5}$

E $\frac{41}{25}$

WORKED SOLUTION

For the whole GP, $a/(1-r) = 8/5$.

The even-numbered terms form a GP with first term ar and ratio r^2 , so $ar/(1-r^2) = 3/5$.

But $a = (8/5)(1-r)$. Substitute into the second equation:

$$[(8/5)(1-r)r] / [(1-r)(1+r)] = 3/5.$$

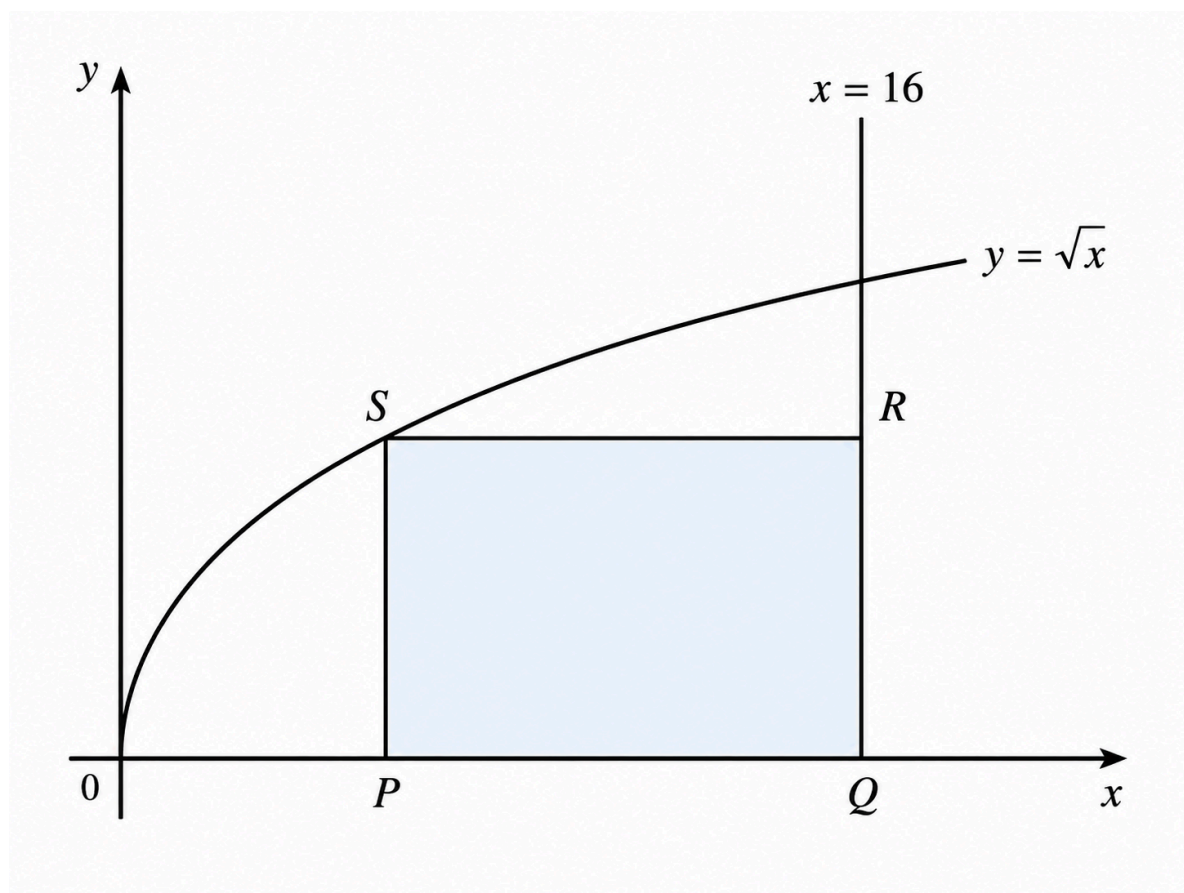
This simplifies to $(8r/5)/(1+r) = 3/5$.

So $8r = 3 + 3r$, hence $5r = 3$ and $r = 3/5$.

Then $a = (8/5)(1-3/5) = (8/5)(2/5) = 16/25$.

Therefore $a + r = 16/25 + 15/25 = \mathbf{31/25}$.

QUESTION 25



PQRS is a rectangle. P and Q lie on the x-axis, Q and R lie on the line $x = 16$, and S lies on the curve $y = \sqrt{x}$. What is the maximum possible area of the rectangle?

A $64\sqrt{3/9}$

B $128\sqrt{3/9}$

C $64\sqrt{3/3}$

D $128\sqrt{3/3}$

E $32\sqrt{3}$

WORKED SOLUTION

Let $S = (x, \sqrt{x})$. Then $P = (x, 0)$, $Q = (16, 0)$, and $R = (16, \sqrt{x})$.

The width of the rectangle is $16 - x$ and its height is $\sqrt{x}$, so

$$A = (16 - x)\sqrt{x}.$$

Let $t = \sqrt{x}$, so $x = t^2$. Then $A = (16 - t^2)t = 16t - t^3$.

Differentiate: $dA/dt = 16 - 3t^2$.

Set to zero: $16 - 3t^2 = 0 \Rightarrow t = 4/\sqrt{3}$.

Then $A_{\max} = 16 \cdot (4/\sqrt{3}) - (4/\sqrt{3})^3 = 64/\sqrt{3} - 64/(3\sqrt{3}) = 128/(3\sqrt{3}) = 128\sqrt{3}/9$.

M2-26

Maximum gradient

Answer E

QUESTION 26

Find the maximum value of the gradient of the curve $y = 1 - 2x + 4x^{3/2} - x^2$, where $x > 0$.

A $\frac{1}{2}$ **B** 1**C** $\frac{3}{2}$ **D** 2**E** $\frac{5}{2}$ **WORKED SOLUTION**

The gradient is $dy/dx = -2 + 6\sqrt{x} - 2x$.

Let $t = \sqrt{x}$, where $t > 0$. Then the gradient becomes

$$g(t) = -2 + 6t - 2t^2 = -2(t^2 - 3t + 1).$$

This is a downward-opening quadratic, so its maximum occurs at $t = -b/(2a) =$

$3/2$.

Then $g_{\max} = -2 + 6 \cdot (3/2) - 2 \cdot (9/4) = -2 + 9 - 9/2 = 5/2$.

M2-27

Trigonometry in triangles

Answer D

QUESTION 27

The internal angles of a triangle are α , β and θ , and

$$3 \tan \alpha - 2 \sin \beta = 2,$$

$$5 \tan \alpha + 6 \sin \beta = 8.$$

What is the value of θ in degrees?

A 15

B 45

C 75

D 105

E 135

WORKED SOLUTION

Let $u = \tan \alpha$ and $v = \sin \beta$. Then we have

$$3u - 2v = 2 \text{ and } 5u + 6v = 8.$$

Multiply the first by 3: $9u - 6v = 6$.

Add to the second: $14u = 14$, so $u = 1$.

Then $3(1) - 2v = 2$, so $v = 1/2$.

Hence $\tan \alpha = 1$. Since α is an internal angle of a triangle, $\alpha = 45^\circ$.

Also $\sin \beta = 1/2$, so an internal angle β could initially be 30° or 150° . But $\beta = 150^\circ$

is impossible because $\alpha + \beta$ would then be 195° , greater than 180° . Thus $\beta = 30^\circ$.
Therefore $\theta = 180^\circ - 45^\circ - 30^\circ = \mathbf{105^\circ}$.

Biology

27 QUESTIONS • COMPLETE SOLUTIONS

Preserved Biology Harder than ESAT source set. Question wording, option order, answer and solution method are unchanged.

B-01

Cells • surface area to volume

Answer C**QUESTION 1**

Two unicellular organisms, X and Y, are approximately cube-shaped.

The side length of Y is twice the side length of X.

Both organisms have:

- the same oxygen demand per unit volume;
- the same maximum rate of oxygen diffusion per unit surface area.

Compared with X, by what factor is the **oxygen demand per unit surface area** of Y greater?

A 1/2**B** 1**C** 2**D** 4**E** 8**WORKED SOLUTION**

For a cube, volume is s^3 and surface area is $6s^2$.

$$\text{oxygen demand per unit surface area} \propto s^3 / 6s^2 = s/6$$

Doubling the side length therefore doubles the oxygen demand per unit surface area.

Answer: 2.

B-02

Movement across membranes

Answer **F**

QUESTION 2

A section of dialysis tubing contains a solution of glucose and starch. It is placed in a large beaker of pure water.

The membrane is permeable to water and glucose, but not to starch.

Consider the statements while the system is approaching equilibrium.

1. While the glucose concentration remains higher inside the tubing than outside, there is a net movement of glucose out of the tubing.
2. Starch can leave the tubing by osmosis because its concentration is higher inside.
3. Even after the glucose concentrations have become equal, water may still show a net movement into the tubing because starch remains inside.

Which statements are correct?

A none

B 1 only

C 2 only

D 3 only

E 1 and 2 only

F

1 and 3 only

G

2 and 3 only

H

1, 2 and 3

WORKED SOLUTION

Glucose can cross the membrane, so while its concentration is higher inside, its net movement is outward by diffusion.

Osmosis describes the net movement of **water**, not starch, so statement 2 is false.

Because starch cannot cross the membrane, it can continue to lower the water potential inside the tubing. Water may therefore continue to enter even after glucose has equilibrated.

Statements 1 and 3 only are correct.

B-03

Diffusion and active transport

Answer B**QUESTION 3**

A root-hair cell contains nitrate ions at a concentration of **4 mmol dm⁻³**.

A respiratory poison is added which completely stops active transport, but the cell membrane remains permeable to nitrate ions.

The cell is placed separately into three soil solutions:

soil solution	nitrate concentration / mmol dm ⁻³
X	1
Y	4
Z	10

Which row gives the expected **net passive movement of nitrate ions immediately after active transport stops?**

A X: into cell; Y: no net movement; Z: out of cell

B X: out of cell; Y: no net movement; Z: into cell

C X: out of cell; Y: into cell; Z: into cell

D X: no net movement; Y: no net movement; Z: into cell

E X: out of cell; Y: out of cell; Z: into cell

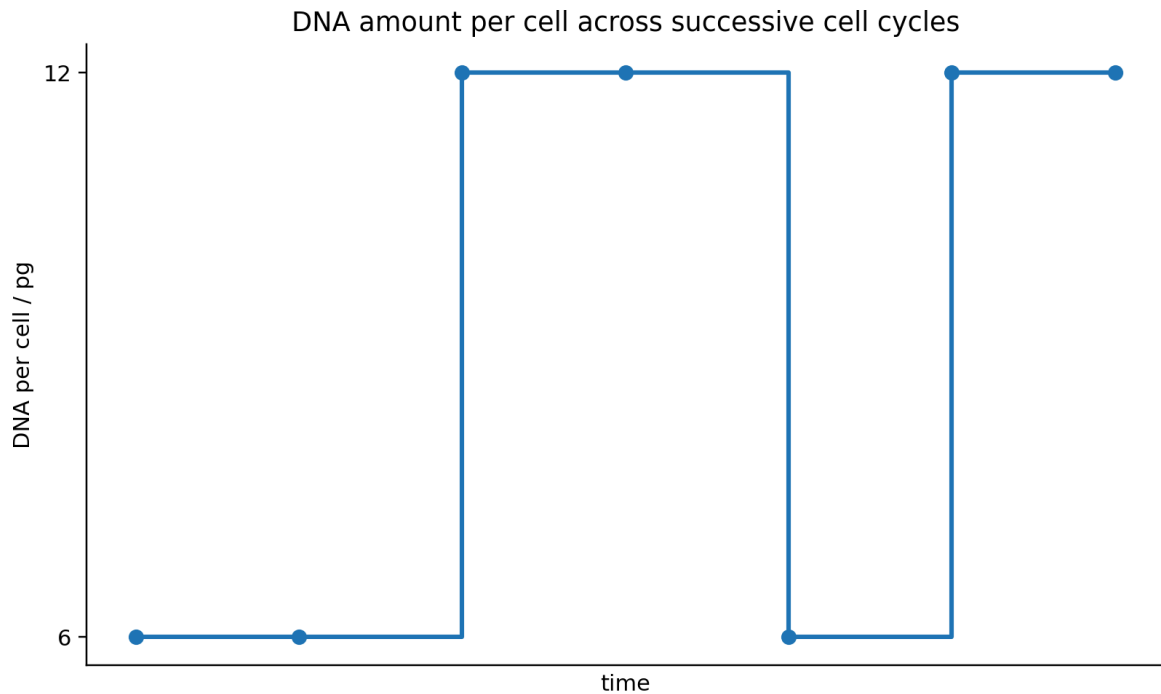
WORKED SOLUTION

With active transport disabled, only passive movement down the concentration gradient remains.

- X: 1 outside versus 4 inside → net movement **out**.
- Y: 4 outside versus 4 inside → **no net movement**.
- Z: 10 outside versus 4 inside → net movement **in**.

Answer B.

QUESTION 4



A diploid species has **12 chromosomes** in each body cell.

Immediately after mitosis, one cell contains **6 pg** of DNA.

It passes through another cell cycle, as represented by the first rise and fall on the graph. Both daughter cells then replicate their DNA and are observed at the **second high plateau**, immediately before their next mitosis.

Which row is correct at this second high plateau?

- A** 1 cell; 12 total chromosomes; 12 pg total DNA
- B** 2 cells; 24 total chromosomes; 12 pg total DNA
- C** 2 cells; 24 total chromosomes; 24 pg total DNA
- D** 2 cells; 48 total chromosomes; 24 pg total DNA

E 4 cells; 48 total chromosomes; 24 pg total DNA

WORKED SOLUTION

After the first mitosis there are two cells. Each has 12 chromosomes and 6 pg DNA.

DNA replication doubles DNA quantity but does not change chromosome number.

So immediately before the next mitosis, each of the two cells has 12 chromosomes and 12 pg DNA.

across both cells: 24 chromosomes and 24 pg DNA

Answer C.

B-05

Meiosis and sex determination

Answer C

QUESTION 5

A male mammal has a diploid chromosome number of $2n = 10$ and an XY sex-chromosome system.

One diploid germ cell completes meiosis normally and produces four sperm.

Across **all four sperm together**, which row is correct?

A 8 total autosomes; 4 total sex chromosomes; X:Y sperm = 2:2

B 16 total autosomes; 2 total sex chromosomes; X:Y sperm = 1:1

C 16 total autosomes; 4 total sex chromosomes; X:Y sperm = 2:2

D 20 total autosomes; 4 total sex chromosomes; X:Y sperm = 1:3

E 16 total autosomes; 8 total sex chromosomes; X:Y sperm = 2:2

WORKED SOLUTION

Each sperm is haploid and therefore contains 5 chromosomes: 4 autosomes plus 1 sex chromosome.

$$4 \text{ sperm} \times 4 \text{ autosomes} = 16 \text{ autosomes}; 4 \text{ sperm} \times 1 \text{ sex chromosome} = 4 \text{ sex chromosomes}$$

Normal meiosis of an XY germ cell gives equal numbers of X-bearing and Y-bearing sperm, so the ratio is 2:2.

Answer C.

B-06

Inheritance · conditional probability

Answer D

QUESTION 6

In an animal species:

- allele A is dominant and produces black fur in genotype Aa;
- genotype aa produces white fur;
- genotype AA is lethal before birth.

Two **living black animals** mate.

What is the probability that **exactly one of their next two live offspring is white**?

A 1/9

B 2/9

C 1/3

D 4/9

E 2/3

WORKED SOLUTION

A living black animal cannot be AA because AA is lethal, so both parents are Aa. Before lethality the cross gives 1/4 AA, 1/2 Aa and 1/4 aa. Among **live** offspring, the probabilities therefore become:

$$P(\text{black} \mid \text{alive}) = 2/3 \quad P(\text{white} \mid \text{alive}) = 1/3$$

Exactly one white among two live offspring can occur in either order:

$$2 \times (1/3) \times (2/3) = 4/9$$

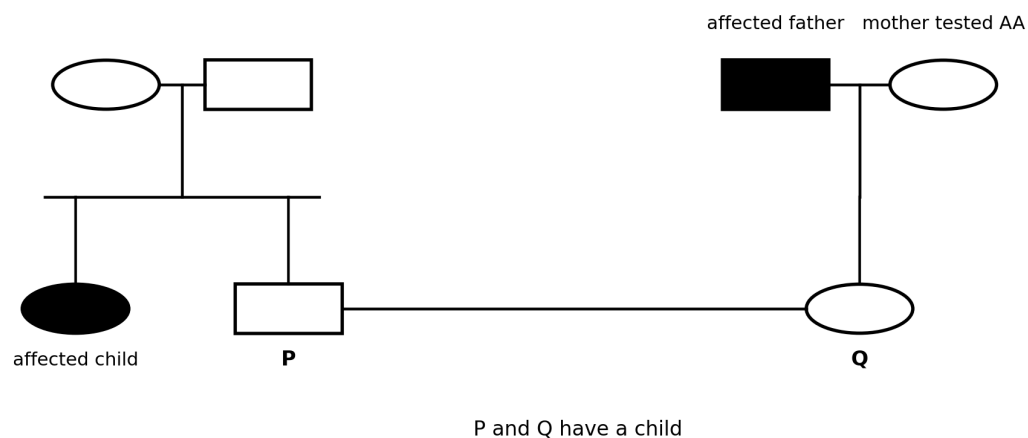
Answer D.

B-07

Pedigrees · conditional probability

Answer C

QUESTION 7



Filled symbol = affected by the same autosomal recessive condition

The same autosomal recessive condition occurs in both families shown.

In the family on the left, two unaffected parents have an affected child and an unaffected son, P.

In the family on the right, Q's father is affected and Q's unaffected mother has been genetically tested and is homozygous dominant.

P and Q have a child.

What is the probability that their child is affected?

A 1/12

B 1/8

C 1/6

D 1/4

E 1/3

WORKED SOLUTION

P's parents must both be Aa because they are unaffected but have an affected aa child.

Given that P is unaffected, P has a $2/3$ probability of being Aa.

Q's father is aa and Q's mother is AA, so Q must be Aa.

If P is a carrier, $Aa \times Aa$ gives an affected child with probability $1/4$.

$$(2/3) \times (1/4) = 1/6$$

Answer C.

B-08

DNA base pairing

Answer D

QUESTION 8

One strand of a DNA molecule has the following base composition:

base	percentage
A	18
T	30
C	22
G	30

The complete double-stranded DNA molecule contains **2000 bases**.

How many **guanine bases** are present in the complete molecule?

A 440

B 480

C 500

D 520

E 600

WORKED SOLUTION

On the complementary strand, the percentage of G equals the percentage of C on the first strand, so it is 22%.

Across the two equal-length strands, the overall fraction of G is therefore:

$$(30\% + 22\%) / 2 = 26\%$$

$$0.26 \times 2000 = 520$$

Answer D.

QUESTION 9

The following coding-DNA triplets have the stated meanings.

triplet	amino acid
GCT	alanine
GCC	alanine
GTT	valine
GAT	aspartic acid
CCT	proline

A gene originally contains **GCT-GAT-CCT**.

Two mutations occur separately:

X: GCC-GAT-CCT

Y: GTT-GAT-CCT

1. Mutation X must alter the amino-acid sequence.
2. Mutation Y alters one amino acid in the encoded sequence.
3. Mutation Y must necessarily produce a visible change in phenotype.

Which statements are correct?

A none

B 1 only

C 2 only

D 3 only

E 1 and 2 only

F 1 and 3 only

G 2 and 3 only

H 1, 2 and 3

WORKED SOLUTION

GCT and GCC both code for alanine, so mutation X does not alter this amino acid.

GTT codes for valine rather than alanine, so mutation Y changes one amino acid.

An amino-acid change does not guarantee a visible phenotypic change.

Statement 2 only is correct — answer C.

B-10

Gene technologies

Answer B

QUESTION 10

A bacterial strain is initially sensitive to kanamycin.

Scientists use a plasmid containing a functional kanamycin-resistance gene. The plasmid is cut at a site **outside** this resistance gene.

A gene of interest is cut with the same restriction enzyme, mixed with the plasmid and DNA ligase, and the bacteria are transformed.

The bacteria are then grown on agar containing kanamycin. Some colonies grow.

Assume that no new mutation conferring kanamycin resistance occurs during the experiment.

Which conclusion is justified?

A Every growing colony must contain the gene of interest.

B Every growing colony must have obtained DNA carrying a functional kanamycin-resistance gene, but growth alone does not show whether the gene of interest was inserted.

C Every bacterium exposed to the plasmid must have been transformed.

D The gene of interest must have entered the bacterial chromosome rather than a plasmid.

E DNA ligase must have cut the plasmid DNA.

WORKED SOLUTION

The original bacteria are kanamycin-sensitive. A colony that grows on kanamycin must therefore have acquired functioning resistance under the conditions described.

However, a cut plasmid can potentially re-form without receiving the desired insert. Kanamycin survival therefore does not prove that the gene of interest is present.

Answer B.

B-11

Stem cells

Answer B

QUESTION 11

Three human cell populations have these properties.

population	observed ability
P	can give rise to all cell types required to form a complete organism
Q	can differentiate into any body-cell type but cannot by itself form a complete organism
R	obtained from adult bone marrow and can form several types of blood cell

A patient requires replacement blood cells.

Which row is correct?

- A** P pluripotent; Q totipotent; R multipotent; least broad sufficient population = P
- B** P totipotent; Q pluripotent; R multipotent; least broad sufficient population = R
- C** P totipotent; Q multipotent; R pluripotent; least broad sufficient population = Q
- D** P pluripotent; Q multipotent; R totipotent; least broad sufficient population = R
- E** P multipotent; Q pluripotent; R totipotent; least broad sufficient population = P

WORKED SOLUTION

P is totipotent because it can generate all cell types needed for a complete organism.

Q is pluripotent because it can generate all body-cell types but not a complete organism.

R is multipotent because it can form a restricted range of related cell types.

For replacement blood cells, R is the least developmentally broad population that could be sufficient.

Answer B.

B-12

Variation · genes and environment

Answer E

QUESTION 12

Two genetically different clonal lines, A and B, are grown under two nitrate conditions. All other measured conditions are kept the same.

clonal line	low nitrate: mean height / cm	high nitrate: mean height / cm
A	12	20
B	16	24

1. Environmental conditions can affect height.
2. Genetic differences could contribute to differences in height.
3. These data prove that nitrate availability is the only environmental factor that can affect plant height.

Which conclusions are supported?

A none

B 1 only

C 2 only

D 3 only

E 1 and 2 only

F 1 and 3 only

G 2 and 3 only

H 1, 2 and 3

WORKED SOLUTION

Within each clonal line, height changes when nitrate availability changes, supporting an environmental effect.

Clonal line B is taller than clonal line A under both nitrate conditions, which is consistent with a genetic contribution.

The experiment does not establish that nitrate is the only environmental factor capable of affecting height.

Statements 1 and 2 only are supported — answer E.

B-13

Enzymes · experimental interpretation

Answer **F**

QUESTION 13

An enzyme is tested at three temperatures and three pH values. The table shows the time required to produce the **same fixed quantity of product**.

	20°C	40°C	60°C
pH 4	180 s	60 s	>300 s
pH 6	120 s	30 s	240 s
pH 8	200 s	75 s	>300 s

1. Of the conditions tested, pH 6 and 40°C gives the highest reaction rate.
2. These results prove that the enzyme's exact optimum is pH 6 and exactly 40°C.
3. At pH 6, the relative reaction rate at 40°C is four times that at 20°C.

Which statements are correct?

A none

B 1 only

C 2 only

D 3 only

E 1 and 2 only

F 1 and 3 only

G 2 and 3 only

H 1, 2 and 3

WORKED SOLUTION

For a fixed quantity of product, rate is inversely proportional to the time taken.

The shortest measured time is 30 s at pH 6 and 40°C, so statement 1 is true.

$$\text{relative rate ratio} = (1/30) / (1/120) = 4$$

Statement 3 is therefore true.

The exact optimum could lie between the values tested, so statement 2 is not justified.

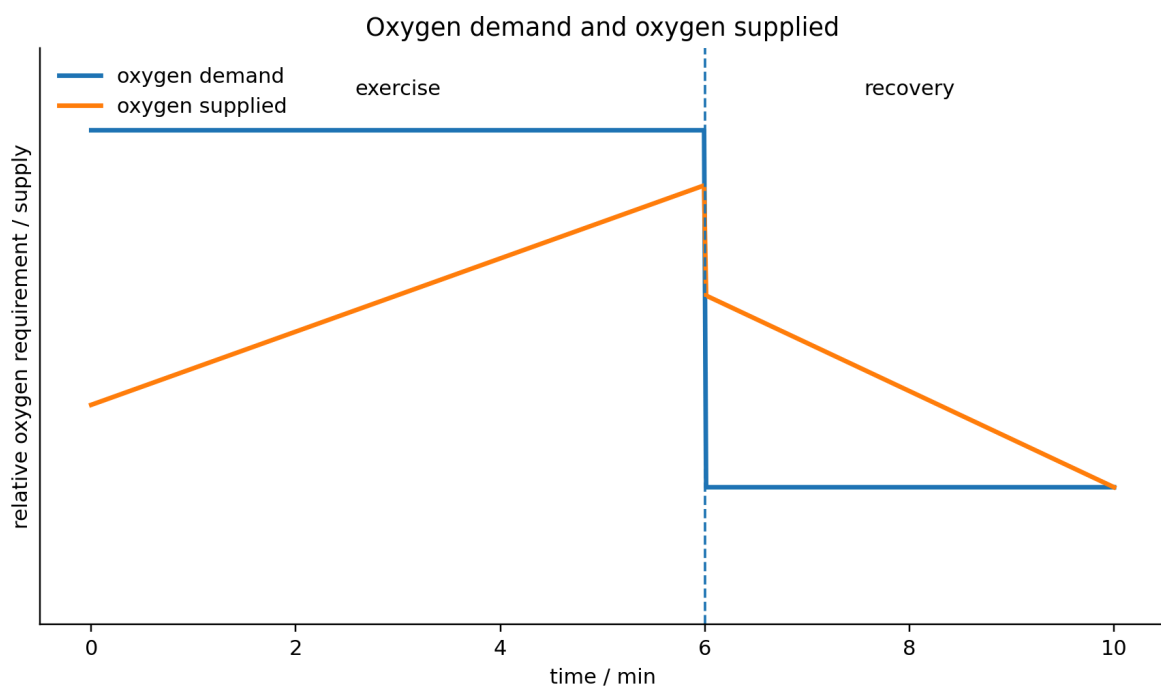
Answer F: 1 and 3 only.

B-14

Respiration during exercise

Answer E

QUESTION 14



The graph shows relative oxygen demand and oxygen supplied during six minutes of exercise followed by recovery.

1. During the exercise period, the graph is consistent with anaerobic respiration contributing to energy release.
2. During the early recovery period, oxygen supply exceeding the current oxygen demand is consistent with repayment of oxygen debt.
3. When oxygen supply and demand become equal at the end of recovery, aerobic respiration must have stopped.

Which statements are correct?

A none

B 1 only

C 2 only

D 3 only

E 1 and 2 only

F 1 and 3 only

G 2 and 3 only

H 1, 2 and 3

WORKED SOLUTION

During exercise, oxygen demand exceeds the oxygen supplied, so aerobic respiration alone cannot account for the full stated demand. This is consistent with anaerobic respiration contributing.

During early recovery, oxygen supply remains above the current demand, which is consistent with repayment of oxygen debt.

Equal supply and demand at the end does not mean respiration has stopped; it means oxygen supply is meeting the current requirement.

Statements 1 and 2 only — answer E.

B-15

Gas exchange · quantitative diffusion

Answer C

QUESTION 15

A simple cubic organism has side length s mm.

Its cells consume oxygen at a rate equivalent to **1 unit per mm^3 per minute**.

Oxygen can cross the entire external surface at a maximum rate of **0.5 units per mm^2 per minute**.

The organism has no internal transport system.

What is the largest value of s for which diffusion across the surface could supply enough oxygen?

A 1 mm

B 2 mm

C 3 mm

D 4 mm

E 6 mm

WORKED SOLUTION

Oxygen demand equals the volume:

$$\text{demand} = s^3$$

Surface area is $6s^2$, so maximum supply is:

$$0.5 \times 6s^2 = 3s^2$$

Enough oxygen is supplied when $3s^2 \geq s^3$. For $s > 0$ this gives $s \leq 3$.

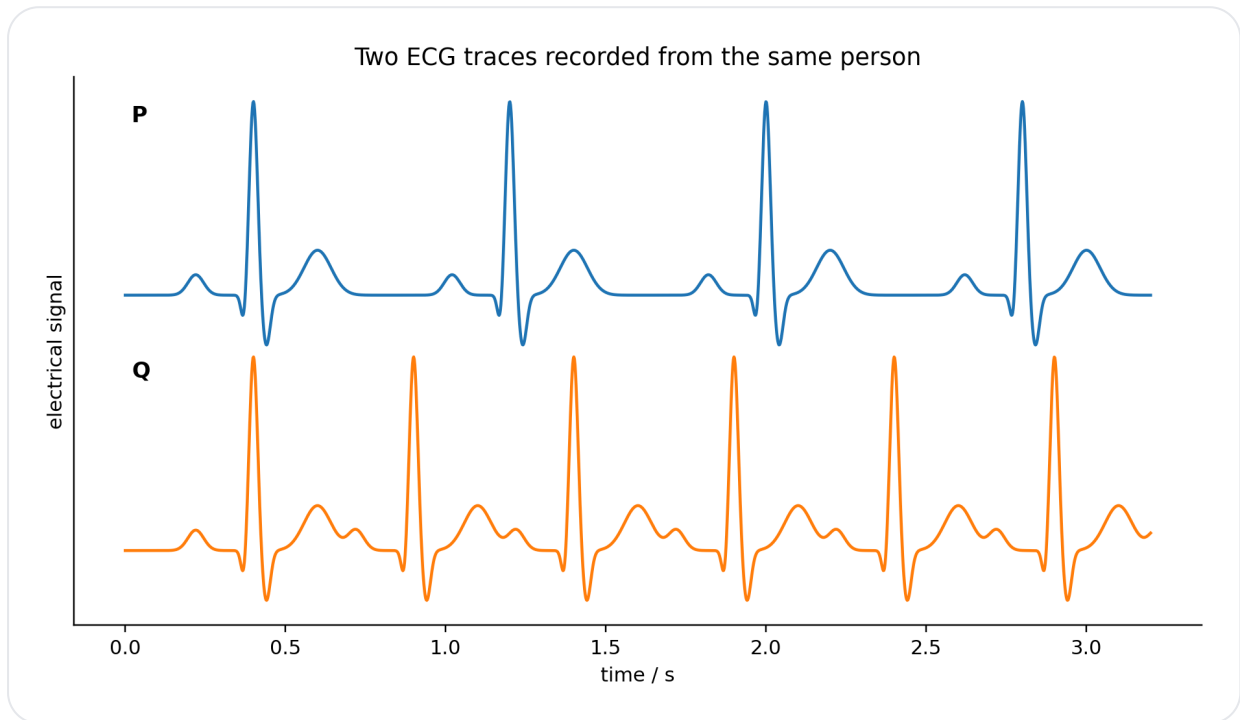
Largest possible side length = 3 mm — answer C.

B-16

ECG · heart rate and cardiac output

Answer **D**

QUESTION 16



Two ECG traces are recorded from the same person.

The spacing between corresponding large peaks is approximately **0.80 s in P** and **0.50 s in Q**.

Assume the volume of blood pumped per heartbeat is unchanged.

Which row is correct?

A P = 60 min⁻¹; Q = 75 min⁻¹; blood/minute ratio Q:P = 1.25

B P = 75 min⁻¹; Q = 100 min⁻¹; ratio = 1.33

C P = 75 min⁻¹; Q = 120 min⁻¹; ratio = 1.50

D P = 75 min⁻¹; Q = 120 min⁻¹; ratio = 1.60

E $P = 80 \text{ min}^{-1}$; $Q = 120 \text{ min}^{-1}$; ratio = 1.50

WORKED SOLUTION

$$P \text{ heart rate} = 60 / 0.80 = 75 \text{ beats min}^{-1}$$

$$Q \text{ heart rate} = 60 / 0.50 = 120 \text{ beats min}^{-1}$$

If stroke volume is unchanged, blood pumped per minute changes in the same ratio:

$$120 / 75 = 1.60$$

Answer D.

B-17

Kidney · blood composition

Answer B

QUESTION 17

The concentrations of four substances are measured in two vessels connected to the same organ.

substance	vessel P	vessel Q
oxygen	9	6
carbon dioxide	5	7
urea	7	2
glucose	5	5

Values are in arbitrary but comparable units.

Which interpretation best fits the data?

A The organ is a lung; P is a pulmonary vein and Q a pulmonary artery.

B The organ is a kidney; P could be a renal artery and Q a renal vein.

- C** The organ is a liver; P must be a hepatic vein and Q a hepatic artery.
- D** The organ is a kidney; P must be the renal vein and Q the renal artery.
- E** The organ is skeletal muscle; P must be a vein and Q an artery.

WORKED SOLUTION

From P to Q, oxygen falls and carbon dioxide rises, consistent with respiration by the organ's cells.

Urea falls sharply, indicating removal from the blood. This strongly fits a kidney.

Blood enters a kidney in a renal artery and leaves through a renal vein, so P could be the renal artery and Q the renal vein.

Answer B.

B-18

Digestion · bile and lipase

Answer B

QUESTION 18

Equal amounts of lipid and lipase are used in four experiments.

tube	lipid	bile	pH	time to fixed endpoint
A	large droplets	absent	8	120 s
B	mechanically emulsified into small droplets	absent	8	45 s
C	large droplets	present	8	50 s
D	large droplets	present	2	>300 s

Which conclusion is best supported?

- A** Bile is itself the enzyme that hydrolyses lipid.

B In these experiments, an important effect of bile could be increasing lipid surface area, because mechanical emulsification produces a similar increase in rate.

C Lipase has its optimum at pH 2.

D Bile converts lipid directly into glucose.

E Lipase functions only when bile is present.

WORKED SOLUTION

Tubes B and C reach the endpoint in very similar times even though B has no bile. In B the lipid has instead been mechanically broken into small droplets.

This supports the idea that an important effect of bile is emulsification, which increases the lipid surface area available to lipase.

Tube D also shows that an unsuitable low pH greatly reduces the reaction rate.

Answer B.

B-19

Homeostasis · ADH

Answer E

QUESTION 19

A drug prevents ADH from increasing the water permeability of the collecting ducts but does not directly affect filtration or glucose reabsorption.

A dehydrated person takes the drug.

Compared with a healthy dehydrated person, which changes are expected?

1. A greater urine volume.
2. A lower concentration of urea in the urine is likely because more water remains in the urine.
3. Glucose must appear in the urine because ADH is absent from the collecting duct.

A none

B 1 only

C 2 only

D 3 only

E 1 and 2 only

F 1 and 3 only

G 2 and 3 only

H 1, 2 and 3

WORKED SOLUTION

ADH normally increases water permeability of the collecting ducts, increasing water reabsorption.

If that effect is blocked, less water is reabsorbed, so urine volume is greater and dissolved urea is likely to be less concentrated.

ADH is not the mechanism responsible for glucose reabsorption, so glucose does not have to appear in the urine.

Statements 1 and 2 only — answer E.

B-20

Nervous and hormonal coordination

Answer **E**

QUESTION 20

A person's hand touches a very hot surface.

The hand is rapidly withdrawn. A few seconds later, adrenaline contributes to an increased heart rate.

1. Withdrawal of the hand can be coordinated through a reflex pathway involving a relay neurone in the central nervous system.
2. Neurotransmitter molecules can diffuse across synapses in the reflex pathway.
3. Adrenaline travels from an endocrine gland to the heart along motor neurones.

Which statements are correct?

A none

B 1 only

C 2 only

D 3 only

E 1 and 2 only

F 1 and 3 only

G 2 and 3 only

H 1, 2 and 3

WORKED SOLUTION

A rapid withdrawal reflex can involve a relay neurone in the central nervous system.

At chemical synapses, neurotransmitter molecules diffuse across the synaptic gap.

Adrenaline is a hormone and is transported in the blood, not along motor neurones.

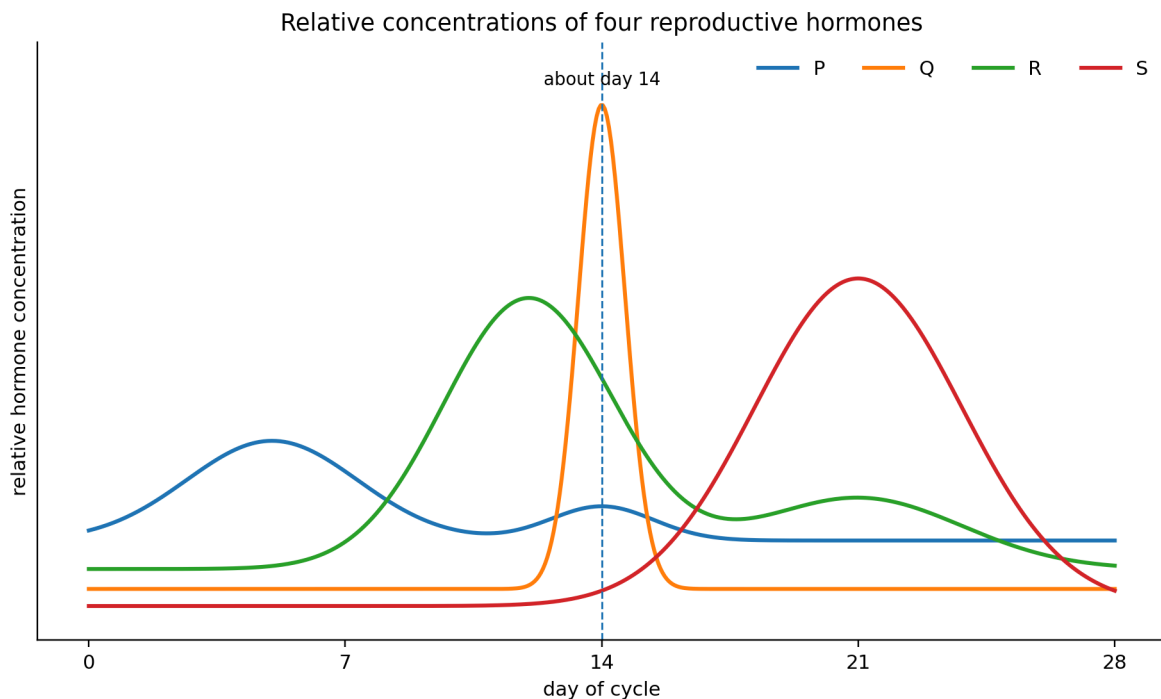
Statements 1 and 2 only — answer E.

B-21

Reproductive hormones

Answer A

QUESTION 21



The graph shows four hormones, P, Q, R and S.

The large narrow peak around day 14 represents the hormone that triggers ovulation.
The broad peak during the second half of the cycle represents the hormone that maintains the uterine lining.

Which pair of curves represents the **two pituitary hormones whose production tends to be inhibited when oestrogen and progesterone are maintained at high levels by combined hormonal contraception?**

A P and Q

B P and R

C Q and S

D R and S

E Q and R

WORKED SOLUTION

The sharp mid-cycle peak is LH, so Q is LH.

The broad second-half peak is progesterone, so S is progesterone.

The remaining pituitary hormone is FSH, represented by P.

The two pituitary reproductive hormones are therefore P and Q.

Answer A.

B-22

Clinical trials · evidence

Answer E

QUESTION 22

A medicine and a placebo are tested on two similar groups.

group	people tested	people whose symptoms improved	people reporting side effects
medicine	200	130	20
placebo	200	90	8

1. The improvement rate is 20 percentage points higher in the medicine group.
2. The proportion reporting side effects is 2.5 times as high in the medicine group as in the placebo group.
3. The trial proves that every improvement in the medicine group was caused by the medicine.

Which are correct?

A none

B 1 only

C 2 only

D 3 only

E 1 and 2 only

F 1 and 3 only

G 2 and 3 only

H 1, 2 and 3

WORKED SOLUTION

medicine improvement = $130/200 = 65\%$

placebo improvement = $90/200 = 45\%$

The difference is 20 percentage points.

side-effect ratio = $(20/200) / (8/200) = 10\% / 4\% = 2.5$

However, improvement also occurred in the placebo group, so the trial does not show that every individual improvement in the medicine group was caused by the medicine.

Statements 1 and 2 only — answer E.

B-23

Ecology · quadrat sampling

Answer **D**

QUESTION 23

Ten randomly positioned **0.50 m × 0.50 m** quadrats are used to sample a plant population.

quadrat	count
1	0
2	4
3	7
4	0
5	5
6	2
7	8
8	0
9	3
10	1

The habitat has an area of **100 m²**.

Which row gives both the estimated population and the **frequency of occurrence** of the plant?

A 300; 70%

B 600; 30%

C 1200; 30%

D 1200; 70%

E 4800; 70%

WORKED SOLUTION

Total organisms counted = 30, so the mean is 3 per quadrat.

Each quadrat has area 0.25 m², giving a density of:

$$3 / 0.25 = 12 \text{ plants m}^{-2}$$

$$\text{estimated population} = 12 \times 100 = 1200$$

The species occurs in 7 of the 10 quadrats, so its frequency is 70%.

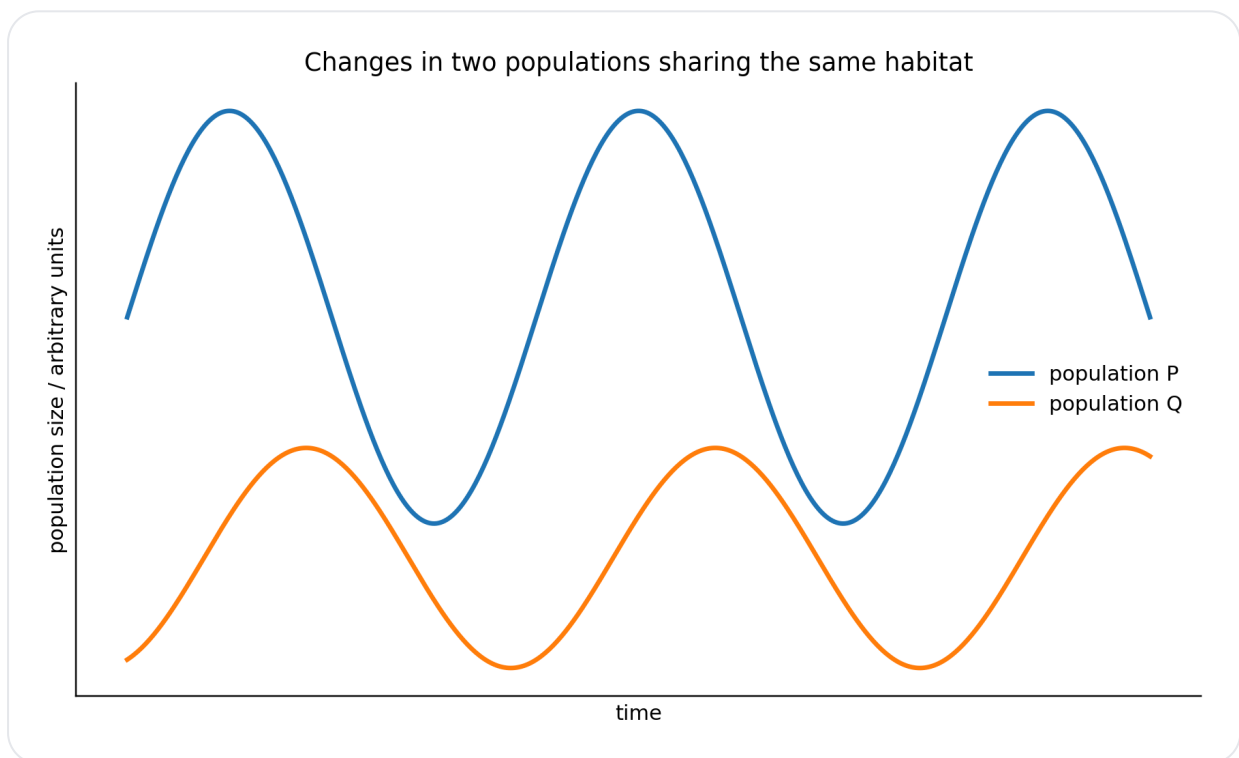
Answer D.

B-24

Ecology · population interactions

Answer F

QUESTION 24



Two populations in the same habitat show the changes in the graph.

Population Q tends to peak after population P.

1. The pattern is consistent with Q being a predator whose population partly depends on P as prey.
2. The graph alone proves that changes in P are the sole cause of changes in Q.
3. If P is an important prey species for Q, a decline in P could later contribute to a decline in Q.

Which statements are justified?

A none

B 1 only

C 2 only

D 3 only

E 1 and 2 only

F 1 and 3 only

G 2 and 3 only

H 1, 2 and 3

WORKED SOLUTION

The delayed peaks are consistent with a possible predator-prey relationship.

If P is important prey, reduced P availability could later reduce Q.

However, the graph alone cannot prove that P is the sole cause of changes in Q because other biotic or abiotic factors may also affect Q.

Statements 1 and 3 only — answer F.

B-25

Carbon cycle · decomposition

Answer E

QUESTION 25

Equal masses of leaf litter containing radioactive carbon, ^{14}C , are placed in three sealed experimental systems.

The amount of $^{14}\text{CO}_2$ released after one week is measured.

system	conditions	$^{14}\text{CO}_2$ produced / arbitrary units
A	non-sterile soil, 25°C	60
B	sterilised soil, 25°C	5
C	non-sterile soil, 5°C	20

1. Living decomposers contribute substantially to carbon release from the litter.
2. Lower temperature can reduce the rate at which carbon is returned from the litter as carbon dioxide.
3. These data show that light is essential for decomposition.

Which conclusions are supported?

A none

B 1 only

C 2 only

D 3 only

E 1 and 2 only

F 1 and 3 only

G 2 and 3 only

H 1, 2 and 3

WORKED SOLUTION

A produces far more labelled CO_2 than sterilised B, supporting a major role for living decomposers.

A and C both contain decomposers, but the lower-temperature system releases much less labelled CO_2 , supporting a temperature effect.

Light was not manipulated, so statement 3 is not supported.

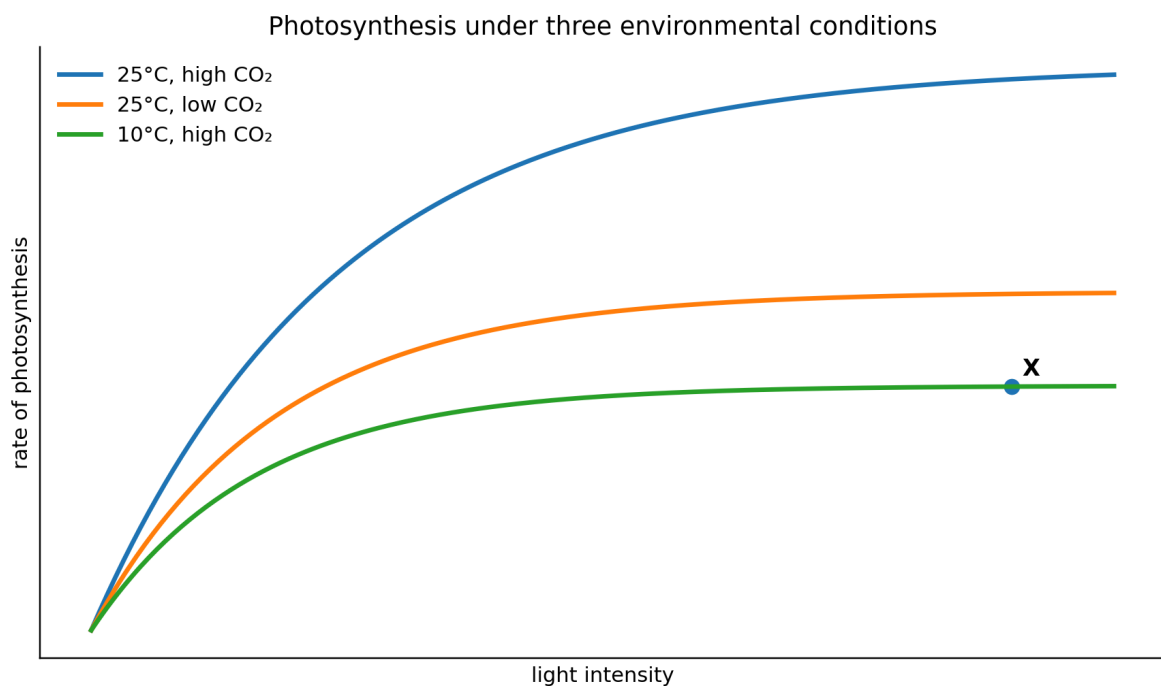
Statements 1 and 2 only — answer E.

B-26

Photosynthesis · limiting factors

Answer C

QUESTION 26



The graph shows photosynthesis under three conditions.

Point X is at high light intensity, high carbon dioxide concentration and **10°C**.

Which single change is most strongly supported by the graph as likely to increase the rate at X?

- A** Increase the light intensity further.
- B** Increase the carbon dioxide concentration further.
- C** Increase the temperature from 10°C towards 25°C.
- D** Decrease the carbon dioxide concentration.

E Decrease the temperature.

WORKED SOLUTION

At X, the 10°C high-CO₂ curve is already near its plateau, so increasing light further is unlikely to have much effect.

Carbon dioxide is already high.

At the same high light and high carbon dioxide concentration, the 25°C curve reaches a much higher rate. The graph therefore supports temperature as the key limiting factor at X.

Answer C.

B-27

Plant physiology · photosynthesis and translocation

Answer H

QUESTION 27

A mature leaf is supplied with radioactive carbon dioxide, ¹⁴CO₂, for a short period.

In one plant, a ring of tissue is removed from the stem below the leaf. The treatment removes the **phloem but leaves the xylem intact**.

A third plant has an intact stem but its treated leaf is kept in darkness.

treatment	radioactivity in sugars in/above treated leaf	radioactivity in roots	relative water uptake
illuminated, intact stem	1000	400	6
illuminated, phloem removed	1300	30	6
dark leaf, intact stem	50	10	5

1. The much greater incorporation of ¹⁴C in illuminated leaves is consistent with carbon dioxide being incorporated into sugars during photosynthesis.
2. The large reduction in radioactivity reaching the roots when phloem is removed supports the role of phloem in sugar transport.

3. The maintenance of water uptake after phloem removal is consistent with xylem remaining functional.

Which conclusions are supported?

A none

B 1 only

C 2 only

D 3 only

E 1 and 2 only

F 1 and 3 only

G 2 and 3 only

H 1, 2 and 3

WORKED SOLUTION

The illuminated intact leaf incorporates far more labelled carbon into sugars than the dark leaf, consistent with carbon dioxide fixation during photosynthesis.

Removing phloem reduces root radioactivity from 400 to 30 while labelled sugar accumulates in or above the treated region, supporting phloem translocation.

Water uptake remains at 6 after phloem removal, consistent with the intact xylem continuing to transport water.

All three conclusions are supported — answer H.

source sets. This solution book is ready to support the later construction of the matching full test.